The Singer conjecture in dimension four

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Abstract

The Singer conjecture says that on the universal cover of a closed aspherical manifold, the L^2-Betti numbers of the cover sit only in the middle dimension. In dimension four the only open case is degree one. We prove the degree-one vanishing, first for finite connected aspherical integral Poincare complexes of formal dimension four (any orientation character) and hence for all closed connected aspherical topological four-manifolds, smoothable or not.

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1 What we are proving

Background: what a manifold is, and what "aspherical" means

A *closed n-manifold* is a compact space without boundary that locally looks like $\mathbb{R}^n$. A *CW complex* is a space built by gluing together cells of various dimensions; every manifold that is not too pathological admits such a description. A space is *aspherical* if its universal cover is contractible. The familiar example is the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$: its universal cover is $\mathbb{R}^n$, which can be squashed to a point. On the other hand the sphere S^2 is *not* aspherical: its universal cover is itself, and S^2 is not contractible.

A manifold (or CW complex) that is aspherical is completely determined in homotopy theory by its fundamental group $\Gamma = \pi_1(M)$: the cover is contractible, so all of the topology of M can be read off from the group action on that contractible space. This is why aspherical manifolds are interesting: their geometry lives in the group.

What L^2-Betti numbers are (informally)

Ordinary Betti numbers $b_p(M)$ count the independent "p-dimensional holes" of a space M. When M has an infinite fundamental group, its universal cover $M \to \widetilde{M}$ is an infinite space, and its Betti numbers are infinite. The trick is to *normalize* them by the size of the group, using the "von Neumann dimension", which is a notion of dimension for spaces with a group action that is allowed to be a fraction. The result $b_p^{(2)}(\widetilde{M})$ is a nonnegative real number, finite for finite CW complexes with finite Betti numbers of the group over $\mathbb{Z}$. On the torus T^2, for example, $b_0^{(2)} = b_2^{(2)} = 0$ and $b_1^{(2)} = 1$; the Euler characteristic is the alternating sum, $\chi(T^2) = 0$, which agrees with $b_0^{(2)} - b_1^{(2)} + b_2^{(2)} = -1 + 1 = 0$. One can show for every finite CW complex that $\chi(M) = \sum_p (-1)^p b_p^{(2)}(\widetilde{M})$.

The Singer conjecture

Conjecture 1.1 (Singer, as formulated for aspherical manifolds)**.** *Let M be a closed aspherical n-manifold with infinite fundamental group. Then*

$$b_p^{(2)}(\widetilde{M}) = 0 \qquad \text{for every } p \neq n/2.$$

The point is that only the middle dimension survives. For a torus T^{2k} this tells us $b_k^{(2)} = \binom{2k}{k}$ and all other L^2-Betti numbers vanish; for a hyperbolic surface it says $b_1^{(2)} = 1$ and the rest vanish, which is what the Gauss-Bonnet theorem predicts. The conjecture was proposed by Singer and was suggested to Dodziuk in the special case of nonpositive curvature. It remains open in most dimensions.

What we prove in dimension four

In dimension four, Poincare duality already forces $b_4^{(2)} = b_0^{(2)}$ and $b_3^{(2)} = b_1^{(2)}$. The degree-zero vanishing ($b_0^{(2)} = 0$ for infinite groups) is elementary, so the only outstanding degree is $p = 1$. Our goal is to prove $b_1^{(2)} = 0$. Equivalently:

Theorem 1.2 (Main theorem)**.** *Let Q be a finite connected aspherical integral Poincare complex of formal dimension four, with arbitrary orientation character. Then*

$$b_p^{(2)}(\widetilde{Q}) = 0 \qquad \text{for all } p \geq 0,\ p \neq 2.$$

The class of complexes is slightly larger than manifolds; see Definition 3.1 below.

Corollary 1.3 (Singer in dimension four)**.** *Let M be a closed connected aspherical topological four-manifold. Then*

$$b_p^{(2)}(\widetilde{M}) = 0 \qquad \text{for all } p \neq 2.$$

Neither orientability nor a smooth structure is needed. In particular the corollary covers non-smoothable four-manifolds, which are topological spaces that admit no PL triangulation or smooth structure.

Why these statements matter

Combined with the L^2 Euler Poincare formula $\chi(M) = \sum_p (-1)^p b_p^{(2)}(\widetilde{M})$, Theorem 1.2 implies $\chi(M) = b_2^{(2)}(\widetilde{M}) \geq 0$ for every closed aspherical four-manifold. Corollary 12.1 records this. Combined with the topological L^2-signature theorem it also yields the Euler–signature inequality $\chi(M) \geq |\sigma(M)|$, where $\sigma(M)$ is the ordinary signature; see Corollary 12.2. Together these force strong restrictions on the possible Betti numbers (and the intersection form) of a closed aspherical four-manifold.

Prior work in low dimensions

The three-dimensional case was handled by Lott and Luck (1995) via three-manifold geometrization. In dimension four, several special cases were known:

- Right-angled Coxeter groups whose nerve is a flag triangulation of S^3 (Davis–Okun, 2001).
- Groups of type VF with a hierarchy, and arbitrary spherical Coxeter groups (Okun–Schreve, 2016).
- Products of surfaces (via Gromov's restriction on Kähler groups of positive first L^2-Betti number).
- Aspherical complex surfaces with residually finite π_1, and those outside class VII_0^+ without that hypothesis (Albanese–Di Cerbo–Lombardi, 2025).
- Even-dimensional Gromov–Thurston branched covers of hyperbolic manifolds (Avramidi–Okun–Schreve, 2025).

Our proof is different in kind: it uses only integral Poincare duality (through §3), the geometry of finite-energy functions, and topological (Cech) cohomology on the resulting boundary. It is not an extension or a combination of the cited arguments.

2 The strategy in one page

Suppose toward contradiction that $b_1^{(2)}(\widetilde{Q}) > 0$. We will derive a contradiction in six stages.

Stage 1 (algebra). Translate the assumption into geometry. The first L^2-Betti number is the dimension of a Hilbert space $\mathcal{D} \ominus \mathcal{D}_0$, where $\mathcal{D}$ is a space of square-summable edge gradients and $\mathcal{D}_0$ is the closure of the finite-support gradients. If $b_1^{(2)} > 0$ then $\mathcal{D}_0$ is a *proper* subspace of $\mathcal{D}$, so there exists a finite-energy function whose gradient is genuinely square-summable and not approximable by finitely supported pieces. This gives us a rich family of finite-energy functions on the Cayley graph. See Section 3.

Stage 2 (compactification). Using this family, build a compact metrizable space $X = \Gamma \sqcup B$: the group Γ is a discrete subset and B is a closed "boundary at infinity". The construction uses Royden-style compactification by bounded finite-energy functions. Add extra coordinates that record which connected component of a coordinate box a point lies in. This ensures a *near-path property*: if two vertices approach the same point of B, they can be joined by a graph path staying close to that point. See Section 4.

Stage 3 (dynamics). Use square-summability of edge lengths in X to prove that the action of Γ on B is a *convergence action*: every escaping sequence of group elements has a subsequence that collapses compact sets off b to a point a. In particular every infinite-order element is parabolic or loxodromic, and B has no global fixed point. The Γ-action is minimal: every orbit is dense. See Section 5.

Stage 4 (boundary topology). Show that the covering dimension of B is at most one, and that Cech cohomology $\check{H}^1(B; \mathbb{F}_2)$ vanishes. The dimension bound uses that "level cuts" of finite-energy functions have small accumulation sets in B. The cohomology vanishing uses integral Poincare duality for the group and shrinks a finite-cover cocycle to a finite-support cellular cocycle, then corrects it and reads off a Čech coboundary. See Section 6.

Stage 5 (cut points). A *cut point* of B is a point whose removal disconnects it. If B has no cut points, set $L = B$; the topology of L is what we need. Otherwise use Bowditch's theory of pretrees: the cut points of B form a simplicial tree, and there is a "leaf block" L (formalized below) with the property

$$L \text{ is a nondegenerate continuum without cut points,} \qquad \dim L \leq 1, \qquad \check{H}^1(L; \mathbb{F}_2) = 0.$$

Moreover L is a retract of B and comes with a graph structure whose vertices are dense in L and have infinite orbit and stabilizer structure. See Sections 7–8.

Stage 6 (path pushing). Suppose L were not locally connected. Then some point x of L would have a neighborhood that contains points y that cannot be connected to x within any smaller neighborhood. We push graph paths toward the boundary using *annuli* (nested pairs of closed sets) and a *three-pointer majority* argument. The key point is *uniformity*: one depth threshold works for every path simultaneously. This forces L to be locally connected, contradiction. See Sections 9–10.

Final contradiction. A nondegenerate compact metric space that is locally connected, has covering dimension ≤ 1, and has no embedded circle (forced by $\check{H}^1(\cdot; \mathbb{F}_2) = 0$) is called a *dendrite*. Every nondegenerate dendrite has a cut point. But L has no cut points by construction. Contradiction. See Section 11.

Therefore our original assumption $b_1^{(2)} > 0$ was false, so $b_1^{(2)}(\widetilde{Q}) = 0$, and by Poincare duality all non-middle L^2-Betti numbers vanish.

3 Topology and algebra: setting up the group

Let $k = \mathbb{F}_2$ throughout the boundary argument. Fix a finite connected aspherical integral Poincare complex Q of formal dimension four, with orientation character $w : \pi_1(Q) \to \{1, -1\}$, and write

$\Gamma = \pi_1(Q)$. We do not fix a particular model; we only use that Q is a finite CW complex satisfying Poincare duality.

Definition 3.1 (Poincare complex, formal dimension n)**.** A finite connected CW complex Q is a *Poincare complex of formal dimension* n if there is an orientation character $w : \pi_1(Q) \to \{1, -1\}$ and a class $[Q] \in H_n(Q; \mathbb{Z}^w)$ (where $\mathbb{Z}^w$ is the local system twisting by w) such that cap product with $[Q]$ gives an isomorphism

$$[Q] \cap - : H^i(Q; V) \longrightarrow H_{n-i}(Q; w \otimes V)$$

for every local coefficient system V. The "$\mathbb{Z}^w$" twist only matters when Q is nonorientable; for orientable Q we can take $w \equiv 1$. When $H_*(Q; \mathbb{Z}^w)$ vanishes outside the fundamental class, the complex is an *integral* Poincare complex.

What the topological input gives us

The strategy is: use only the algebraic shadow of Poincare duality, plus the fact that asphericity makes Γ tractable. We collect those consequences now.

Lemma 3.2 (Fundamental group of a PD4 complex)**.** *The group Γ is infinite, finitely presented, and torsion-free. Moreover, Q is homotopy equivalent to a finite four-dimensional CW complex, and the augmented cellular chain complex of that model is a finitely generated free $\mathbb{Z}\Gamma$-resolution of $\mathbb{Z}$ of length four.*

Proof. Wall's dimension theorem [1] provides a four-dimensional CW complex homotopy equivalent to Q; because Q is finite, we may take the model finite (the reduced finiteness obstruction is zero for a space already finitely dominated). Asphericity plus finite four-dimensional model means the model's universal cover is contractible, so its cellular chains give a free resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length four.

Torsion-free. Suppose $g \in \Gamma$ has prime order p. Let $C = \langle g \rangle \cong \mathbb{Z}/p$. Restrict the resolution just described to C. We still have a free $\mathbb{Z}C$-resolution of $\mathbb{Z}$ of finite length ≤ 4. But a cyclic group of prime order has 2-periodic cohomology: $H^{2j}(C; \mathbb{Z}) \cong \mathbb{Z}/p$ for every $j \geq 1$. A resolution of length ≤ 4 cannot possibly give nonzero cohomology in arbitrarily high degrees, so we arrive at a contradiction. Hence no finite-order element exists.

Infinite. Suppose Γ finite. By torsion-freeness, Γ is trivial, so Q is simply connected and aspherical, hence contractible. But Poincare duality over k gives $H_4(Q; k) \cong H^0(Q; k) \cong k$, contradicting contractibility (a point has no H_4). So Γ is infinite.

Finitely presented. A finite four-dimensional CW model has finitely many 1-cells and 2-cells, so its fundamental group is finitely presented. □

Lemma 3.3 (Manifolds reduce to Poincare complexes)**.** *Every closed connected aspherical topological four-manifold M is homotopy equivalent to a finite connected aspherical integral Poincare complex of formal dimension four.*

Proof. A compact topological manifold is metrizable and is an absolute neighborhood retract (ANR): the coordinate neighborhood of any point is an ANR, and Hanner's locality theorem upgrades this to a global ANR; see [2, Thms. 3.2–3.3]. Its covering dimension is ≤ 4 by the closed-sum theory of dimension; see [7, Theorems 1.1.2, 1.5.3, 1.7.7, 1.8.2]. West's homotopy dimension estimate then supplies a finite four-dimensional simplicial complex F homotopy equivalent to M; see [3, Cor. 5.4]. This F is not a triangulation of M, just a homotopy model.

Transport the fundamental class and orientation character of M to F. Topological Poincare duality with local coefficients gives cap-product isomorphisms for every coefficient system on M: [4, Theorem 1, with $R = \mathbb{Z}$], and see also [5, Thm. 10.2, Rem. 10.3] and [6, Thm. A.16]. Transporting along the homotopy equivalence gives F the PD property. Combined with asphericity of M, this makes F a finite connected aspherical integral Poincare complex of formal dimension four. □

What Poincare duality gives us algebraically

The two key algebraic facts are that the group-ring module $\mathbb{Z}\Gamma$ (and its $k\Gamma$ analogue) has vanishing first and second cohomology, and that one can read off good information about subgroups of Γ through Shapiro's lemma.

Lemma 3.4 (Group-cohomological vanishing from duality)**.** *With $k = \mathbb{F}_2$, the group Γ satisfies*

$$H^i(\Gamma; k\Gamma) = 0 \quad (i = 1, 2).$$

Moreover, for every subgroup $J \leq \Gamma$ there is a natural isomorphism

$$H^1(\Gamma; k[\Gamma/J]) \cong H_3(J; k).$$

In particular this group vanishes if J is finite or two-ended.

Proof. The orientation twist of a PD4 complex is invisible over k. Because Q is aspherical,

$$H^i(Q; V) \cong H^i(\Gamma; V)$$

for every local system V. Apply Poincare duality (which, over k, does not see the orientation character) with V a k-vector space:

$$H^i(\Gamma; V) \cong H_{4-i}(\Gamma; V), \quad 0 \leq i \leq 4.$$

Take $V = k\Gamma$ to get $H^i(\Gamma; k\Gamma) \cong H_{4-i}(\Gamma; k\Gamma)$. Taking $i = 1$ gives $H^1 \cong H_3$, and $i = 2$ gives $H^2 \cong H_2$. The chain complex of Γ with coefficients in $k\Gamma$ is just the augmented $k\Gamma$-resolution of k tensored with $k\Gamma$, and since that resolution is finite of length four, $H_p(\Gamma; k\Gamma) = 0$ for $p > 0$ (a free resolution of k is exact in positive degrees). Therefore $H^1(\Gamma; k\Gamma) = H_3(\Gamma; k\Gamma) = 0$, and $H^2(\Gamma; k\Gamma) = 0$.

The Shapiro isomorphism. The permutation module $k[\Gamma/J] = k\Gamma \otimes_{kJ} k$. If R_* is a free right $k\Gamma$-resolution of k then

$$R_* \otimes_{k\Gamma} (k\Gamma \otimes_{kJ} k) \cong (R_*|_J) \otimes_{kJ} k,$$

which is a free kJ-resolution of k. Thus $H_j(\Gamma; k[\Gamma/J]) \cong H_j(J; k)$. Combined with $H^i(\Gamma; V) \cong H_{4-i}(\Gamma; V)$, this gives (with $i = 1$) $H^1(\Gamma; k[\Gamma/J]) \cong H_3(J; k)$.

Vanishing for small J. If J is finite, it is trivial by Lemma 3.2, so $H_3(J; k) = 0$. If J is two-ended, it has a finite normal subgroup N with quotient J/N infinite cyclic or infinite dihedral; this is [1, Lemma 4.1]. Torsion-freeness forces N trivial, so J is infinite cyclic (the dihedral case is impossible because dihedral groups contain involutions). But an infinite cyclic group is an iterated extension of the trivial group by $\mathbb{Z}$, so its integral homology is concentrated in degrees 0 and 1; over k the same holds, so $H_3(J; k) = 0$. □

Lemma 3.5 (Group is one-ended)**.** *Γ is one-ended: removing any finite set of vertices of a Cayley graph leaves exactly one infinite connected component.*

Proof. Choose a finite presentation of Γ and let $\mathcal{C}$ be its Cayley two-complex (a simply connected 2-complex on which Γ acts freely). The corresponding Cayley graph G is locally finite, since the presentation is finite. The augmented cellular complex of $\mathcal{C}$ is exact in degrees zero and one; extending by free modules in higher degrees gives a free resolution in which each cohomology is the same as in $\mathcal{C}$.

Suppose some finite vertex set, when removed from G, leaves two distinct infinite components. Let $f : \Gamma \to k$ be the indicator of one of those infinite components, with value 0 on the deleted vertices. Its gradient df is a finitely supported edge cochain (because the graph is locally finite and f is nonzero only on one component), and df is a cellular cocycle (its coboundary vanishes around every cell because f is $\{0,1\}$-valued; there are no 2-cell coefficients outside k). By Lemma 3.4, $H^1(\Gamma; k\Gamma) = 0$, so the cohomology class of df in the group-ring cocycle complex is a coboundary: $df = du$ for a *finitely supported* vertex function u. Then $f - u$ is a locally constant function on Γ, so $f - u$ is constant (the group is connected). But f equals 1 on an infinite set and 0 on another infinite set, whereas u has finite support; this is impossible.

The infinite connected locally finite Cayley graph has at least one end (that is standard: greedily choose longer and longer escaping paths), and we have just excluded two or more. Hence Γ is one-ended. □

No splittings over finite or two-ended subgroups

The path-pushing argument uses Bowditch's theory of tree actions. We need to know Γ does not split over a finite or two-ended group.

Lemma 3.6 (Splitting obstruction)**.** *Consider a Γ-action on a simplicial tree without inversions.*

(i) *If the action has no fixed vertex, then the edge stabilizers cannot all be finite or two-ended.*

(ii) *If the action is minimal and the tree has an edge, then every edge stabilizer is neither finite nor two-ended.*

Consequently Γ has no nontrivial splitting over a finite or two-ended subgroup.

Proof. Fix a base vertex o and an edge orbit E with stabilizer J. Identify E with Γ/J. For $g \in \Gamma$ let $c(g) \in k$ be the mod-two number of edges of E on the geodesic $[o, go]$ projected through E (equivalently, the mod-two intersection number of $[o, go]$ with E). Cancellation of edges traversed twice gives the cocycle identity

$$c(gh) = c(g) + g\,c(h).$$

If J is finite or two-ended, the Shapiro-based vanishing (Lemma 3.4) makes c principal: $c(g) = gv - v$ for some $v \in k[\Gamma/J]$. Hence the coefficient size of c is bounded:

$$|\,[o, go] \cap E\,| \le 2|\operatorname{supp} v| \qquad (g \in \Gamma). \tag{3.1}$$

Part (i). Let S be a finite generating set of Γ. The finitely many paths $[o, so]$, $s \in S$, together contain only finitely many edges. Translates of those paths cover every $[o, go]$ (write g as a word in S and cancel backtracking). Therefore only finitely many edge orbits occur on the paths $[o, go]$, $g \in \Gamma$. An action without a fixed vertex has unbounded vertex orbits: if the orbit of some vertex were bounded, its convex hull would give an invariant center (a fixed vertex when there are no inversions). So one of the finitely many edge-orbit crossing counts $|\,[o, go] \cap E\,|$ is unbounded. If all edge stabilizers were finite or two-ended, this would contradict (3.1).

Part (ii). Fix an edge orbit E and collapse each connected component of the union of edges outside E. The result is a nontrivial simplicial tree without inversions, on which Γ acts. It has no fixed vertex: the preimage of a fixed vertex would be a proper nonempty invariant subtree of the original. Its orbits are therefore unbounded, and distance in this quotient counts exactly the edges of E on the original path $[o, go]$. Again (3.1) rules out a finite or two-ended stabilizer.

Applying this to the Bass–Serre tree of any splitting shows that Γ does not split over a finite or two-ended subgroup. □

Reducing the L^2 problem to $b_1^{(2)} = 0$

We now recall the algebraic meaning of $b_1^{(2)}$ and how Poincare duality reduces the whole vanishing statement to just degree one.

Lemma 3.7 (Only degree one is missing)**.** *For the fixed PD4 complex Q,*

$$b_0^{(2)}(\widetilde{Q}) = b_4^{(2)}(\widetilde{Q}) = 0, \qquad b_3^{(2)}(\widetilde{Q}) = b_1^{(2)}(\widetilde{Q}),$$

and $b_p^{(2)}(\widetilde{Q}) = 0$ for $p > 4$. Consequently Theorem 1.2 reduces to proving $b_1^{(2)}(\widetilde{Q}) = 0$.

Proof. By Lemma 3.2, Q has a finite four-dimensional cellular model, so there are no cellular chains above degree four and $b_p^{(2)} = 0$ for $p > 4$. A harmonic square-summable 0-cochain on the connected universal cover is constant; since Γ is infinite, that constant must be zero, so $b_0^{(2)} = 0$.

For the duality statement we must handle a nontrivial orientation character. Let $\Gamma^+ \leq \Gamma$ be the kernel of the orientation character $\Gamma \to \{1, -1\}$ and $Q^+ \to Q$ the corresponding two-fold (or one-fold) cover. Its index $d = [\Gamma : \Gamma^+]$ is 1 or 2. Finite covers of Poincare complexes are again Poincare complexes of the same formal dimension [10]. The orientation character of Q^+ is the restriction of w to Γ^+ and is therefore trivial, so Q^+ is an *oriented* finite connected aspherical integral PD4 complex. By Lemma 3.2 pick a finite four-dimensional model F of Q^+.

Transport the fundamental class of Q^+ to F. Poincare duality with local coefficients applies to Q^+, hence to F [1, Lemma 1.1]. In particular, applying duality to the regular group-ring local system shows that cap product defines a quasi-isomorphism

$$C^*(\widetilde{F}; \mathbb{Z}) \xrightarrow{\cong} C_*(\widetilde{F}; \mathbb{Z})$$

(with degree reversal and a $g \mapsto g^{-1}$ twist on the coefficient module). Both sides are bounded complexes of finitely generated free $\mathbb{Z}\Gamma^+$-modules, so the mapping cone is a bounded acyclic complex of free modules. A bounded acyclic complex of free modules that splits at every surjection is contractible (start in the lowest degree and split each surjection onto the kernel already obtained). Hence cap product is a chain homotopy equivalence.

Extending scalars to the group von Neumann algebra $\mathcal{N}(\Gamma^+)$, homotopies remain homotopies, and finite-matrix dimension comparison identifies the von Neumann dimensions of the two complexes. Consequently

$$b_p^{(2)}(\widetilde{Q}; \mathcal{N}(\Gamma^+)) = b_{4-p}^{(2)}(\widetilde{Q}; \mathcal{N}(\Gamma^+)) \quad (0 \leq p \leq 4).$$

Finally, finite-index subgroup scaling [8, Thm. 1.35(9)] gives

$$b_p^{(2)}(\widetilde{Q}; \mathcal{N}(\Gamma^+)) = d \cdot b_p^{(2)}(\widetilde{Q}; \mathcal{N}(\Gamma)).$$

Dividing by d gives $b_p^{(2)} = b_{4-p}^{(2)}$ for Q itself. Taking $p = 0$ gives $b_4^{(2)} = b_0^{(2)} = 0$, and taking $p = 1$ gives $b_3^{(2)} = b_1^{(2)}$. Degree two is the only one not forced to vanish. □

Where the Hilbert space of finite-energy gradients comes from

What we need going forward: a rich supply of finite-energy functions built from the presentation. Assume toward contradiction $b_1^{(2)} > 0$ from now on. Then we can find such a function that is not approximable by finitely supported gradients. Let us set that up.

Choose a finite presentation of Γ with relators of length at most three (subdividing longer relators by new generators achieves this; Tietze transformations preserve all the cohomology we care about). Let $\mathcal{C}$ be the Cayley two-complex with graph G; write $E(G)$ for the edge set. As noted in Lemma 3.2, G is connected and locally finite, $\mathcal{C}$ is simply connected, and each cellular edge belongs to a uniformly bounded number of cell boundaries. We keep parallel edges and loops as separate cellular edges, and choose one orientation on each edge.

For a real function $f : \Gamma \to \mathbb{R}$ and an oriented edge $e = uv$ put $df(e) = f(v) - f(u)$. Define the Hilbert spaces

$$\mathcal{D} = \{df : f : \Gamma \to \mathbb{R}, \sum_{e \in E(G)} |df(e)|^2 < \infty\}, \quad \mathcal{D}_0 = \overline{\{df : f \text{ finitely supported}\}}$$

where the closure is in $\ell^2(E(G); \mathbb{R})$ and the sum counts each cellular edge once.

Lemma 3.8 (Finite-energy gradients and $b_1^{(2)}$)**.** *The space $\mathcal{D}$ is a closed real Hilbert subspace of $\ell^2(E(G); \mathbb{R})$ and $\mathcal{D}_0 \subseteq \mathcal{D}$. If $b_1^{(2)}(\widetilde{Q}) > 0$ then $\mathcal{D}_0 \neq \mathcal{D}$.*

Proof. $\mathcal{D}$ is closed. Write $F(\mathcal{C})$ for the set of two-cells. The cellular coboundary

$$d_1 : \ell^2(E(G); \mathbb{R}) \to \ell^2(F(\mathcal{C}); \mathbb{R})$$

is bounded: each relator has at most three edges (counted with multiplicities), and edge–cell incidence is uniformly bounded by the presentation. An ordinary cellular 1-cocycle on $\mathcal{C}$ is a gradient: integrate along edge paths from a fixed base vertex, and path-independence is exactly the cocycle condition on the simply connected $\mathcal{C}$. Therefore $\mathcal{D} = \ker d_1$, which is closed.

$\mathcal{D}_0$ description. Likewise $d_0 : \ell^2(\Gamma; \mathbb{R}) \to \ell^2(E(G); \mathbb{R})$ is bounded (bounded vertex degree). Since finite-support functions are dense in $\ell^2(\Gamma)$ and d_0 is continuous, the closure of the image of d_0 is the closure of the image of finitely supported functions on $\ell^2(G)$. So $\mathcal{D}_0 = \overline{\operatorname{im} d_0}$; in particular $\mathcal{D}_0 \subseteq \ker d_1 = \mathcal{D}$. Note that this does *not* assert $\operatorname{im} d_0$ itself is closed.

Comparison with $b_1^{(2)}$. The augmented cellular chains of $\mathcal{C}$ are exact in degrees zero and one. Extend to a free resolution by choosing a free module P_3 mapping onto $\ker(\partial_2 : C_2(\mathcal{C}) \to C_1(\mathcal{C}))$, then free modules onto successive kernels. The cellular terms in degrees $0, 1, 2$ are finitely generated free; so the hypothesis of the finite-matrix comparison (below) is met. The singular chains of $\widetilde{Q}$ are free $\mathbb{Z}\Gamma$-modules: a deck transformation fixing a singular simplex fixes its image points and hence is the identity. Thus Lemma 3.9 applies.

Over $\mathbb{C}$, that comparison identifies $b_1^{(2)}(\widetilde{Q})$ with the dimension of

$$\ker d_1 \ominus \overline{\operatorname{im} d_0}.$$

All the matrices involved are real, so this complex Hilbert space is the complexification of $\mathcal{D} \ominus \mathcal{D}_0$. If $\mathcal{D} = \mathcal{D}_0$, its dimension would be zero. Therefore $b_1^{(2)} > 0$ forces strict inclusion. □

Lemma 3.9 (Finite-matrix comparison)**.** *Let Λ be any group and let P_* be a free $\mathbb{Z}\Lambda$-resolution of $\mathbb{Z}$ whose terms in degrees $0, 1, 2$ are finitely generated. Suppose a contractible space Z has a free Λ-action with free singular chains over $\mathbb{Z}\Lambda$. Then*

$$b_1^{(2)}(Z; \mathcal{N}(\Lambda)) = \dim_{\mathcal{N}(\Lambda)}(\ker \partial_1 \cap \ker \partial_2^*)$$

where the operators on the right act on the Hilbert spaces $\ell^2(\Lambda)\otimes_{\mathbb{Z}\Lambda} P_i$. The same space is the first reduced Hilbert cohomology for $d_0=\partial_1^*$ and $d_1=\partial_2^*$.

Proof. Chain-homotopy comparison. The augmented singular chains of Z are a free resolution of $\mathbb{Z}$. The comparison theorem for free resolutions gives chain-homotopy equivalences with P_*. Tensoring with the von Neumann algebra preserves chain homotopies, so it preserves homology, which is what the definition of $b_1^{(2)}$ computes. No flatness of $\mathcal{N}(\Lambda)$ over $\mathbb{Z}\Lambda$ is needed because we only need to preserve homotopy equivalences.

Dimension comparison. For a matrix over $\mathcal{N}(\Lambda)$, the extended von Neumann dimension of the algebraic kernel equals the trace of the projection onto its Hilbert-space kernel. The dimension of the algebraic image (taken as von Neumann dimension) equals the trace of the projection onto the closure of its Hilbert image. By additivity of dimension,

$$\dim H_1(\mathcal{N}\Lambda\otimes P_*)=\dim\bigl(\ker\partial_1\ominus\overline{\operatorname{im}\partial_2}\bigr)=\dim\bigl(\ker\partial_1\cap\ker\partial_2^*\bigr).$$

On the other hand the harmonic representative space for reduced cohomology is

$$\ker d_1\ominus\overline{\operatorname{im} d_0}=\ker\partial_2^*\cap\ker\partial_1.$$

These are the same closed Hilbert submodule. The argument works in every degree with finite free neighbours. This is the comparison underlying [8, Rem. 1.31] and [8, Lemmas 6.51–6.53]. □

From now on: assume toward contradiction $b_1^{(2)}(\widetilde{Q})>0$, so $\mathcal{D}_0\subsetneq\mathcal{D}$. We have obtained a nonzero quotient $\mathcal{D}/\mathcal{D}_0$, together with Γ being finitely presented, torsion-free, one-ended, having no splitting over a finite or two-ended subgroup, and having the group-cohomological vanishings of Lemma 3.4. These are the algebraic inputs for the rest of the proof.

4 Building a compact boundary

This is the heart of the proof's constructive core. Our standing assumption throughout is that $b_1^{(2)}(\widetilde{Q})>0$, equivalently $\mathcal{D}_0\subsetneq\mathcal{D}$. We will use this to build a compact metrizable space $X=\Gamma\sqcup B$ whose "boundary" B is a nice continuum, and to arrange that vertices approaching the same boundary point can be joined by graph paths staying close to that point. That last property is the *near-path property*; it converts membership in B into a concrete connectivity statement about paths in G.

One function with connected level sets

Start with a "starter function" whose level sets (the subgraphs $\{f<t\}$ and $\{f>t\}$) are connected, and which does not tend to a constant at infinity. That function will provide the base coordinate for the compactification.

Lemma 4.1 (A good function exists)**.** *There exists a finite-energy function $f:\Gamma\to[0,1]$ which does not tend to a single constant at infinity, and for which both $\{f<t\}$ and $\{f>t\}$ are connected vertex subgraphs whenever $0<t<1$ is regular for f (i.e. not attained at any vertex). Every such cut $S(f,t)$ is dually connected.*

Proof. Step 1 (the functional). Fix a vertex $a\in\Gamma$. For each $b\neq a$, the functional

$$L_b(df)=f(b)-f(a)$$

is well defined and continuous on $\mathcal{D}$: a finite path of length ℓ from a to b gives $|L_b(df)| \le \sqrt{\ell}\,\|df\|_2$ by Cauchy–Schwarz. The functional is nonzero because if we test it on the gradient of the singleton indicator $\mathbf{1}_{\{b\}}$, we get $\mathbf{1}_{\{b\}}(b) - \mathbf{1}_{\{b\}}(a) = 1 - 0 = 1$. Let $r_b \in \mathcal{D}$ be the Riesz representative: $L_b(dg) = \langle r_b, dg\rangle$.

Step 2 (min-norm representative). The unique gradient of minimal norm satisfying $L_b(df_b) = 1$ is

$$df_b = \frac{r_b}{\|r_b\|_2^2},$$

with potential f_b normalized by $f_b(a) = 0$.

Step 3 (some df_b escapes $\mathcal{D}_0$). The vectors r_b span a dense subspace of $\mathcal{D}$: any dg orthogonal to every r_b would satisfy $L_b(dg) = 0$ for every b, so the potential g is constant, so $dg = 0$. Since $b_1^{(2)} > 0$, we have $\mathcal{D}_0 \subsetneq \mathcal{D}$, so some $df_b \notin \mathcal{D}_0$. Fix such a b and its minimizer df.

Step 4 (normalize to $[0,1]$). The map $\phi(x) = \min(1, \max(0, x))$ satisfies $|\phi(x) - \phi(y)| \le |x - y|$ for all x, y. Replacing f by $\phi \circ f$ preserves the values 0 and 1 at a and b (which were forced by the min-norm condition), cannot increase any edge energy, and yields another candidate minimizer. Uniqueness of the minimizer forces $0 \le f \le 1$ everywhere.

Step 5 (no constant at infinity). Suppose f tends to a constant c outside finite vertex sets. For $\varepsilon > 0$ define

$$u_\varepsilon(v) = \operatorname{sgn}(f(v) - c) \cdot \max(|f(v) - c| - \varepsilon, 0).$$

The support of u_ε is finite by hypothesis (only finitely many vertices satisfy $|f(v) - c| \ge \varepsilon$). The function $(f - c) - u_\varepsilon$ is the truncation of $f - c$ to $[-\varepsilon, \varepsilon]$, so its edge differences are bounded in absolute value by $|df|$ and tend pointwise to zero as $\varepsilon \to 0$. By dominated convergence, $du_\varepsilon \to df$ in ℓ^2. But du_ε has finite support, so $df \in \mathcal{D}_0$, contradiction.

Step 6 (lower level sets connected). Fix a regular $t \in (0, 1)$. Suppose a component U of $\{f > t\}$ misses b. Define $f' = f$ off U, $f' = t$ on U. This preserves $f(a) = 0$ (below t, so $a \notin U$) and $f(b) = 1$ (above t, so $b \notin U$). Internal edge energies inside U drop to zero. On the boundary, some edge $e = uv$ has $u \in U$ and $v \notin U$ (because G is connected and U is a proper subset); on that edge $|df'(e)| < |df(e)|$ strictly. So $\mathcal{E}(f') < \mathcal{E}(f)$, contradicting that f minimizes energy subject to $L_b = 1$. Hence every upper component contains b, and the upper side is connected. The lower side is connected by the identical argument with a and b interchanged.

Step 7 (dual connectedness). Let C be a dual component of $S(f, t)$: a component of the straddling edges under the adjacency "both lie on the boundary of a common 2-cell". Restrict the cochain $d\mathbf{1}_{\{f>t\}}$ to C and set it to zero on all other edges. Consider any cell: the signed sum of this restricted cochain around that cell is either the original zero sum (if the cell's straddling edges are in C) or an entirely zero sum (if they are not, or if some are and some are not, but every crossing edge of the cell lies in one dual component, so all-or-nothing). Simple connectivity of $\mathcal{C}$ gives a potential u for the restricted cochain. That potential is constant on each of the two connected level sides $\{f > t\}$ and $\{f < t\}$, and it has nonzero difference across every edge of C. Consequently, it must have nonzero difference across *every* edge of $S(f, t)$ (otherwise one of those edges would have zero difference, but u would agree with $\mathbf{1}_{\{f>t\}}$ there, contradiction). So $C = S(f, t)$: the wall is dually connected. □

Lemma 4.2 (Coordinate cuts have finitely many dual components)**.** *Every regular cut of a coordinate in $\mathcal{F}_0$ (defined below) has finitely many dual components. A nonempty regular cut of a translate of f is dually connected, and a regular cut of a singleton indicator is finite.*

Proof. Enlarge $\mathcal{F}_0$ by all translates of f and all singleton indicators $\mathbf{1}_{\{v\}}$ (with $v \in \Gamma$). A translate of f also has connected level sides and dually connected cuts (translations of subgroups preserve

the properties of Lemma 4.1). Levels outside $[0,1]$ give empty cuts. A nonempty cut of a singleton indicator is finite: it consists of edges incident to a single vertex, which is finite by local finiteness of G. □

Coordinate boxes: well-behaved regions

We use "boxes" in the base-coordinate cube: finitely many level conditions. "Good" boxes will have finitely many graph components and form a basis for the ambient topology.

Definition 4.3 (Box, good box)**.** A *box* is determined by finitely many conditions

$$O = \{(z_F) : a_i < z_{F_i} < b_i,\ 1 \le i \le m\}, \qquad F_i \in \mathcal{F}_0.$$

Endpoints a_i, b_i are real numbers (possibly outside $[0,1]$). We retain the defining inequalities as part of the box data, and we use the same symbol for its vertex set in G (obtained by evaluating the coordinates). A box is *good* if:

(a) every endpoint a_i, b_i is *regular* for F_i, and

(b) for every ordered pair of endpoints (c_i, c_j) of cuts of coordinates in the box data, the relation $\{(e, e') : e \in S(F_i, c_i),\ e' \in S(F_j, c_j),\ e \sim e'\}$ is finite, where $e \sim e'$ means $e = e'$ or e, e' both lie on a common 2-cell.

Lemma 4.4 (Good boxes are a basis)**.** *Every good box has finitely many graph components on its vertex set. The collection of good boxes forms a countable* Γ*-invariant basis for the base-coordinate cube* $[0,1]^{\mathcal{F}_0}$*.*

Proof. Good boxes have finitely many components. Fix a good box and list its endpoint cuts as $S_1, \dots, S_N$. Delete all of their edges from G, obtaining a graph G'. On every component of G' all endpoint signs are constant.

Let E be the finite set of cut edges appearing in more than one cut, or sharing a cell with an edge of another cut. (We need to know E is finite. This is part of the definition of "good": the intersected relation is finite by hypothesis.) Then $S_i \setminus E$ still has finitely many dual components: bounded degree of the cut adjacency graph plus finite deletion of vertices.

Claim: no two distinct dual components of $S_i \setminus E$ can be incident to a common component of G'. Suppose edges $e_1, e_2 \in S_i \setminus E$ are on a common triangular cell. On the specified side of cut i (both e_1, e_2 have one endpoint below t and one above), either their below-endpoints coincide or are joined by the remaining cell edge. That remaining edge is not in S_i (it's on the specified side), and by the definition of E, it's not in any other endpoint cut either. So the remaining edge is in G'. By induction along a dual path, all endpoints on either specified side of a dual component $D \subset S_i \setminus E$ fall into a single component of G'. Each such dual component is incident to at most two G'-components.

Every edge of E is likewise incident to at most two G'-components (one on each side). So each component of G' is either incident to at least one cut edge (if a cut is nonempty) or is a full component of G. In the first case, bounded finite count of cut edges and their incident components gives finitely many G'-components. In the second, $G' = G$ is connected. So the vertex set of O has exactly the components of the induced subgraph of G' (by constancy of signs on components of G').

Good boxes form a basis. Write $e \sim e'$ when $e = e'$ or e, e' lie on a common cell. The symmetric relation has degree bounded by a constant D: bounded cell size and bounded cell incidence. For

finite-energy coordinates F, F', Tonelli's theorem gives

$$\int_{\mathbb{R}^2} \#\{(e,e') : e \sim e',\ e \in S(F,s),\ e' \in S(F',t)\}\, ds\, dt = \sum_{e \sim e'} |dF(e)| \cdot |dF'(e')|.$$

Cauchy–Schwarz bounds this by $D\|dF\|_2\|dF'\|_2 < \infty$. Thus almost every pair (s,t) has finitely many interacting pairs (e,e'). The same argument works when $F = F'$, since s and t range independently.

For a fixed finite list of box coordinates, intersect the finitely many full-measure conditions above and exclude the countably many vertex values. What remains is a dense set of endpoint tuples in the parameter domain, giving arbitrarily small good boxes about every point of the cube.

Countable invariant basis. There are countably many finite lists of base coordinates. For each, choose a countable dense set of good endpoint tuples. The resulting boxes form a countable basis. Add all their translates: still countable, still a basis, and still good (goodness is Γ-invariant because the action permutes coordinates). □

Fix such an invariant countable basis and call it $\mathcal{O}$.

Coordinates distinguishing box components

A box may have several graph components even though each individual level side is connected (this is possible when several coordinates are combined). We add *localized* coordinates that record which component of a box a vertex lies in.

Definition 4.5 (Localized coordinate)**.** Fix $O \in \mathcal{O}$, with conditions $a_i < F_i < b_i$ for $1 \le i \le m$. Define the *margin function*

$$\psi_O = \max\big(0,\ \min(\{1\} \cup \{F_i - a_i : 1 \le i \le m\} \cup \{b_i - F_i : 1 \le i \le m\})\big).$$

This is continuous in the base coordinates, is Γ-equivariant, takes values in $[0,1]$, and is positive exactly on the interior of O. For each graph component U of O (there are finitely many by Lemma 4.4), define the *localized coordinate*

$$h_{O,U} = \psi_O \cdot \mathbf{1}_U$$

on vertices. Let $\mathcal{F}$ be the family consisting of $\mathcal{F}_0$ and all of these localized coordinates, over all $O \in \mathcal{O}$ and all components U of O.

Lemma 4.6 (Localized coordinates are finite energy with controlled cuts)**.** *Every member of $\mathcal{F}$ has finite energy. For almost every $t \in (0,1)$,*

$$S(h_{O,U}, t) \subseteq \bigcup_{i=1}^{m} \big(S(F_i, a_i + t) \cup S(F_i, b_i - t)\big), \tag{4.1}$$

with every displayed level regular for its coordinate.

Proof. Energy bound. The min of $F_i - a_i$ and $b_i - F_i$ is 1-Lipschitz with respect to each coordinate individually; hence

$$|d\psi_O(e)| \le \max_{1 \le i \le m} |dF_i(e)|$$

for every edge e.

For two adjacent vertices with positive margin, they lie in the same component of O (any connected subgraph of the good box has its vertices in the same component; here vertices with positive margin are inside O and adjacent, so in the same G-component of O). Thus either both contribute the margin to $h_{O,U}$, both contribute zero (both outside U), or the endpoint outside U has zero margin. In every case

$$|dh_{O,U}(e)| \le |d\psi_O(e)|, \qquad \|dh_{O,U}\|_2^2 \le \sum_{i=1}^{m} \|dF_i\|_2^2 < \infty.$$

Shifted walls. Let e be an edge with $h_{O,U}(e^-) < t < h_{O,U}(e^+)$. The endpoint where the value is above t belongs to U (margin strictly positive) and satisfies every $F_i - a_i > t$ and $b_i - F_i > t$. At the other endpoint, either it is still in U but some margin dropped below t (so an edge crosses $F_i = a_i + t$ or $F_i = b_i - t$), or it is outside U with margin zero by adjacency (so again some margin is zero, i.e. again an edge crosses $F_i = a_i$ or $F_i = b_i$). Since $t < 1$, the cap 1 in the definition of ψ_O does not itself create cuts. All the crossed levels are of the form $a_i + t$ or $b_i - t$. Excluding the countably many values of t where these coincide with vertex values, we get the inclusion (4.1). □

The compact space X and its boundary B

We now assemble the compactification. Map each vertex $v \in \Gamma$ to its profile $(F(v))_{F\in\mathcal{F}} \in [0,1]^{\mathcal{F}}$, and let X be the closure of these profiles in the product space $[0,1]^{\mathcal{F}}$. The singleton indicators $\mathbf{1}_{\{v\}}$ in $\mathcal{F}$ make the profile map injective (the $\mathbf{1}_{\{v\}}$ coordinate is 1 only at v), so we identify Γ with its image in X. They also make each vertex isolated in X: the set $\{x \in X : \mathbf{1}_{\{v\}}(x) > 1/2\}$ is open in X and equals $\{v\}$.

Define the *boundary* $B = X \setminus \Gamma$. Since Γ is open in X (each vertex isolated), B is closed in the compact space X, hence compact.

Every coordinate $F \in \mathcal{F}$ extends continuously to X (as the projection). The margin function ψ_O also extends, through its expression in base coordinates. Left translation by Γ acts on X by permuting coordinates; this makes the translation a homeomorphism of X, since X is determined by the invariant family $\mathcal{F}$.

The metric ρ. Enumerate $\mathcal{F} = \{F_1, F_2, F_3, \ldots\}$ and choose $c_j = 2^{-j}/(1 + \|dF_j\|_2) > 0$. Then $\sum_j c_j(1 + \|dF_j\|_2) = \sum_j 2^{-j} = 1 < \infty$. Define

$$\rho(x,y) = \sum_j c_j |F_j(x) - F_j(y)|, \qquad \lambda(e) = \rho(e^-, e^+). \tag{4.2}$$

The sum converges uniformly, so ρ is a continuous metric inducing the product topology on X. By Minkowski's inequality applied to finite partial sums, followed by monotone convergence,

$$\|\lambda\|_{\ell^2(E(G))} \le \sum_j c_j \|dF_j\|_2 < \infty. \tag{4.3}$$

Consequently, for any fixed $\epsilon > 0$, only finitely many edges have ρ-length $\ge \epsilon$ (otherwise an infinite number of edges each contribute ϵ^2, contradicting summability).

Lemma 4.7 (The boundary is a nondegenerate continuum)**.** *The boundary B is a nondegenerate compact connected metric space: it is compact, nonempty, has at least two points, and is connected.*

Proof. Nonempty. A closed subset of X contained entirely in the isolated vertices would be compact and discrete, hence finite. Since Γ is infinite, its closure contains points outside Γ. So $B \neq \varnothing$.

Nondegenerate. If $B = \{x\}$ for a single point x, then for each $\epsilon > 0$ the vertex set $\{v : |f(v) - f(x)| \geq \epsilon\}$ would have to be finite (its closure in X would otherwise meet $B = \{x\}$ at a point other than x, or fail to include x but be nonempty, and any accumulation would be in B). This would make f tend to the constant $f(x)$ at infinity, contradicting Lemma 4.1. So B has at least two points.

Connected. Suppose B is disconnected, so $B = B_1 \sqcup B_2$ with B_1, B_2 nonempty compact disjoint. The distance between them is some $\delta > 0$ (compactness). Choose neighborhoods N_1, N_2 of B_1, B_2 in X whose closures have distance $> \delta/2$ (so they are disjoint) and cover B (since $B = B_1 \sqcup B_2$). The complement $X \setminus (N_1 \cup N_2)$ is a closed subset of X disjoint from B, hence a finite vertex set. Call it F.

Each N_i contains infinitely many vertices (else its intersection with B would be at most finite, contradicting that B_i is nonempty and has no isolated points, since it is a closed subset of the compact metric space X; more concretely, any point of B_i is a limit of vertices of Γ). So infinitely many vertices lie in N_1 and infinitely many in N_2.

Only finitely many graph edges join a vertex of N_1 to a vertex of N_2: if infinitely many did, then infinitely many such edges would have length $\geq \delta/2$, contradicting (4.3). Delete their vertices together with the finite set F. The remaining graph has no path between the two sides: any path from N_1 to N_2 would either stay in $X \setminus (N_1 \cup N_2)$ (impossible: that set is finite and all its vertices are deleted), or cross via one of the finitely many edges between N_1 and N_2 (all their endpoints deleted). So the remaining graph splits into at least two infinite components. A connected locally finite graph minus finitely many vertices has only finitely many components, so we really get infinite components on each side, contradicting one-endedness (Lemma 3.5). □

The near-path property

Connectedness of B is not enough to control paths near a boundary point. We now show that vertices approaching the same boundary point can be joined by graph paths whose vertices also approach that point. This is where the localized coordinates are essential.

Lemma 4.8 (The near-path property)**.** *For every $x \in B$ and every open neighborhood W of x in X, there is an open neighborhood $V \subseteq W$ of x such that any two vertices in V can be joined by a finite graph path whose vertices all lie in W.*

Equivalently: if $u_n, v_n \in \Gamma$ both converge to the same $x \in B$, then there exist joining graph paths P_n with $\sup\{\rho(w, x) : w \text{ is a vertex of } P_n\} \to 0$.

Proof. Component identities. For a good box $O \in \mathcal{O}$ with its finite component list $U_1, \ldots, U_r$, the following identities hold on vertices:

$$\sum_{j=1}^{r} h_{O,U_j} = \psi_O, \qquad h_{O,U_i} h_{O,U_j} = 0 \quad (i \neq j).$$

By continuity they extend to all of X. Consequently, if $\psi_O(x) > 0$, exactly one component $U_O(x)$ satisfies

$$h_{O,U_O(x)}(x) = \psi_O(x) > 0,$$

and all vertices sufficiently near x also fall in this component (by continuity of the localized coordinates and the fact that the $h_{O,U}$ are nonnegative, sum to ψ_O).

Choice of a small feature. Shrink W to a neighborhood defined by finitely many coordinate conditions, each with positive tolerance around its value at x. The base coordinates give a continuous projection $X \to [0,1]^{\mathcal{F}_0}$. Choose a good basis box O' about the base profile of x small enough to have three properties, stated verbatim below as (P1)–(P3):

(P1) every tested base coordinate satisfies its prescribed tolerance throughout O';

(P2) for every tested localized coordinate $h_{O,U}$ with $\psi_O(x) > 0$, we have $O' \subset O$, and the variation of ψ_O over O' is $<$ the prescribed tolerance;

(P3) if a tested box has $\psi_O(x) = 0$, its margin is $<$ the prescribed tolerance throughout O'.

These are finitely many open conditions on the base profile of x, so Lemma 4.4 supplies such a good basis box O'.

Selection of the relevant component. We have $\psi_{O'}(x) > 0$ (the box O' contains the base profile of x). Let $U' = U_{O'}(x)$ be the component selected by $h_{O',U'}(x) = \psi_{O'}(x)$.

For any tested box O with $\psi_O(x) > 0$: since $O' \subset O$ and U' is connected inside O, we have $U' \subseteq$ a single component of O. That component must be $U_O(x)$ because, by density of vertices near x and continuity of coordinates, there is a vertex near x where both $h_{O',U'}$ and $h_{O,U_O(x)}$ are positive; and their product identities force these two positive-mass coordinates to be the same one.

Conclusion for U'. Now consider any tested localized coordinate $h_{O,U}$, evaluated on U':

- If $\psi_O(x) > 0$ and $U = U_O(x)$, then $h_{O,U}|_{U'} = \psi_O|_{U'}$ (since U' is inside the selected component and therefore the indicator $\mathbf{1}_U$ equals 1 there). Its variation is controlled by (P2).
- If $\psi_O(x) > 0$ but $U \neq U_O(x)$, then $h_{O,U} = 0$ identically on U' and at x.
- Finally, if $\psi_O(x) = 0$, then $0 \leq h_{O,U} \leq \psi_O$ everywhere (since $h_{O,U} = \psi_O \mathbf{1}_U$), so the value is controlled by (P3).

Thus all vertices of U' lie in W.

The neighborhood V. Set

$$V = W \cap \{y \in X : h_{O',U'}(y) > \tfrac{1}{2}\psi_{O'}(x)\}.$$

This is open in X, contains x (because $h_{O',U'}(x) = \psi_{O'}(x) > \frac{1}{2}\psi_{O'}(x)$), and every vertex in V belongs to U'. Since U' is a connected component of a subgraph, any two vertices of U' can be joined by a path in U', hence in W. That proves the neighborhood formulation.

Sequential formulation. Apply the neighborhood formulation to successively smaller metric balls $W_k = B_\rho(x, 1/k)$ about x, obtaining V_k with the stated property, and choose radii slowly enough that any prescribed pair of converging sequences $u_n, v_n \to x$ eventually lie inside the same V_k. Conversely, failure of the sequential statement for a fixed W would give two sequences $u_n, v_n \to x$ with no joining paths in W, contradicting the neighborhood formulation. □

We summarize the properties of X, B, and the metric ρ obtained so far:

(i) $X = \Gamma \sqcup B$ is compact metrizable; Γ is discrete (its points are isolated); B is a closed nonempty connected compact subset with at least two points.

(ii) The edge-length function $\lambda(e) = \rho(e^-, e^+)$ is square-summable: $\|\lambda\|_{\ell^2} < \infty$.

(iii) (near-path) For every $x \in B$ and every neighborhood W of x in X, there is a neighborhood $V \subseteq W$ of x such that any two vertices of V can be joined by a graph path inside W.

These three properties are the only role of the compactification until Section 6.

5 Convergence dynamics on the boundary

Now we bring the action of Γ into play. Recall the metric ρ on X and the edge-length function $\lambda(e) = \rho(e^-, e^+)$. From (4.3) we know $\lambda \in \ell^2(E(G))$.

Standing notations: for a function $F : X \to \mathbb{R}$ and a regular level t, $S(F,t)$ is the set of edges whose endpoints lie on opposite sides of t. Two edges are *near* if they are the same edge or occur together on the boundary of a common 2-cell.

Escape to infinity and the two-sided move

First a gentle observation that drives everything else.

Lemma 5.1 (Moving a bounded distance moves us a little)**.** *For every fixed $s \in \Gamma$,*

$$\rho(u, us) \longrightarrow 0 \qquad (u \longrightarrow \infty \text{ in } \Gamma). \tag{5.1}$$

Here "$u \to \infty$" means u escapes every finite subset of Γ. The convergence is uniform over any fixed finite set of choices of s.

Proof. Fix a finite edge path from 1 to s and translate it by u. Each translated edge escapes every finite edge set as $u \to \infty$. Because $\lambda \in \ell^2$, and because ℓ^2 functions tend to zero outside finite sets, each summand $\lambda(ue)$ tends to zero. Summing the finitely many edges of the path gives $\rho(u, us) \le \sum_e \lambda(ue) \to 0$. The same argument works for a fixed finite list of choices of s (the corresponding paths give finitely many edges, and each summand still tends to zero). □

Lengths of translated level sets

The square-summability controls how the level sets of any base coordinate behave after translation.

Lemma 5.2 (Translated cut lengths)**.** *For every base coordinate $F \in \mathcal{F}_0$ and every $g \in \Gamma$,*

$$\int_0^1 \sum_{e \in S(F,t)} \sum_{e' \text{ near } e} \lambda(ge')\, dt = \sum_e |dF(e)| \sum_{e' \text{ near } e} \lambda(ge'). \tag{5.2}$$

This quantity is finite and tends to 0 as $g \to \infty$. There is a constant C, independent of g, F, t, such that for every dually connected set D of edges,

$$\operatorname{diam}_\rho\{gv : v \text{ is an endpoint of an edge of } D\} \le C \sum_{e \in D} \sum_{e' \text{ near } e} \lambda(ge'). \tag{5.3}$$

Moreover the sum of these diameter bounds over distinct dual components is bounded by the same expression with their union in place of D.

Proof. Equality (5.2). For each edge, the set of levels t at which it straddles is an interval of length $|dF(e)|$. Substituting $\sum_{e \in S(F,t)} X_e$ for a summable family (X_e) and applying Tonelli's theorem exchanges the level integral with the edge sum, giving the equality.

Finiteness and vanishing. Define $a(e') = \sum_{e : e' \text{ near } e} |dF(e)|$. Bounded incidence of "near" (uniformly in either variable, because each cell has bounded size and bounded cell incidence) makes $a \in \ell^2(E(G))$ (Cauchy-Schwarz against the ℓ^2 function $e \mapsto |dF(e)|$ over the uniformly bounded neighborhoods). The right side of (5.2) is $\sum_{e'} a(e')\lambda(ge')$, finite by Cauchy-Schwarz. To see it tends to zero as $g \to \infty$, approximate a and λ in ℓ^2 by finitely supported functions: for

finitely supported a_0, λ_0, the pairing $\sum_{e'} a_0(e')\lambda_0(ge')$ vanishes for large g (the supports of a_0 and $g \cdot \mathrm{supp}(\lambda_0)$ are eventually disjoint), and Cauchy-Schwarz controls the error.

Diameter bound (5.3). Fix a dually connected set D and any two edges $e_1, e_2 \in D$. Join them by a simple dual path: a sequence of edges $e_1 = E_0, E_1, \ldots, E_n = e_2$ where consecutive E_i, E_{i+1} lie on a common cell. For each consecutive pair, traverse the boundary of a common cell. This uses each consecutive pair exactly once, and the remaining edges used in the traversal are "near" those in D. Bounded cell size and bounded cell incidence give a uniform bound C' on the number of times any near edge is used. Then by triangle inequality

$$\rho(gu, gv) \le C' \sum_{e \in D} \sum_{e' \text{ near } e} \lambda(ge'),$$

where u, v are any two endpoints of edges in D. Summing over all pairs (or taking the diameter directly) yields (5.3) with $C = C'$. Summing over dual components gives the last claim. □

Lemma 5.3 (Collapse of translated cuts)**.** *Suppose $g_n \to a \in B$. Given finitely many coordinates from $\mathcal{F}$, there is a subsequence such that, for almost every level of each coordinate, all endpoints of the g_n-translated straddling set tend uniformly to a. If also $g_n^{-1} \to b \in B$, the subsequence can be chosen so that the corresponding inverse-translated straddling sets tend uniformly to b.*

Proof. Fix a finite list $F_1, \ldots, F_r$ of base functions. Lemma 5.2 applied to each yields sums $\sum_n \int_0^1 \sum_{e \in S(F_i,t)} \sum_{e' \text{ near } e} \lambda(g_n e')\, dt$. Along a subsequence these are summable in n (any nonnegative summable-in-n-sequence argument works: pick a subsequence making the sums finite). For almost every level, the nonnegative inner sums then tend to zero as $n \to \infty$.

Fix a regular base cut $S(F, t)$ at such a level. It has finitely many dual components (Lemma 4.2). Choose one endpoint v_C in each nonempty dual component. The translates $g_n v_C$ tend to a by Lemma 5.1. The diameter estimate (5.3) with our chosen level makes every endpoint of the translated cut tend to a too (the sum tends to zero).

For a localized coordinate $h_{O,U}$: its straddling set is contained in a finite union of base straddling sets at levels of the form $a_i + t$ or $b_i - t$ (Lemma 4.6). Include those base coordinates in the finite list. Translations and reflections of the level parameter preserve measure zero sets, so almost every t has the required estimates for all the covering cuts. Levels outside $[0, 1]$ contribute empty cuts.

To handle both directions, choose the summable-in-ℓ^2-over-n subsequence simultaneously for g_n and g_n^{-1}. □

The convergence action

Proposition 5.4 (Convergence action)**.** *If $g_n \to a \in B$ and $g_n^{-1} \to b \in B$, then*

$$g_n x \longrightarrow a \quad \textit{uniformly for } x \textit{ in each compact subset of } X \setminus \{b\}. \tag{5.4}$$

Consequently Γ acts on B by a convergence action; that is, on the space of distinct triples of B, the action is properly discontinuous.

Proof. Compact-uniform convergence. Suppose (5.4) fails. After taking a subsequence, there is $x_n \to r \neq b$ with $g_n x_n \to s \neq a$. Since vertices are dense in X, choose $v_n \in \Gamma$ with $\rho(v_n, x_n) < 1/n$ and $\rho(g_n v_n, g_n x_n) < 1/n$. Then $v_n \to r$ and $g_n v_n \to s$.

Choose a coordinate F_1 separating r from b (so $F_1(r) \neq F_1(b)$) and a coordinate F_2 separating s from a. Apply Lemma 5.3 to the pair (F_1, F_2) and to a level strictly between the respective values that avoids the null exceptional sets; call these the first and second cuts.

After the subsequence, the first translated cut has all endpoints uniformly near a, and the second inverse-translated cut has all endpoints uniformly near b.

Let C_n be the graph component of the r-side of the first cut containing v_n (the r-side being the side of the cut where $F_1 > F_1(r)$, say). Note $F_1(b) < t < F_1(r)$, so a small neighborhood of b misses the whole r-side (all its vertices satisfy $F_1 > t$). Both sides have vertices (the cut straddles).

A graph path from v_n to the other side of the cut must contain a first straddling edge; its endpoint w_n on the r-side of that edge lies in C_n. Since the inverse-translated second cut is uniformly near b, none of its edges can have an endpoint on the r-side of the first cut for large n. Since w_n is on the r-side and near a (uniformly), $g_n w_n$ lies on the s-side of the second cut ($s \neq a$ and the second cut separates s from a). But then g_n of a path in C_n from v_n to w_n must include an edge of the second cut (since $g_n v_n$ and $g_n w_n$ sit on opposite sides of it), a contradiction: the inverse-translated second cut has no endpoints on the r-side of the first cut for large n.

Proper discontinuity on triples. Suppose, for contradiction, that infinitely many distinct group elements take some point of one compact set K_1 of distinct triples to some compact set K_2 of distinct triples. Pass to converging subsequences: source triple $(x_1, x_2, x_3) \in K_1$ and target triple $(y_1, y_2, y_3) \in K_2$ and group elements g_n. Passing further, $g_n \to a \in B$ and $g_n^{-1} \to b \in B$ (limits exist by compactness of X; they lie in B since group vertices are isolated). At least two coordinates of the source triple differ from b (only one coordinate can equal b). By (5.4), both of those coordinates map to a in the limit, contradicting that the target triple has distinct entries. □

Parabolic elements and minimality

In a convergence action on a compact metric space with at least three points, every infinite-order element is either *parabolic* (one fixed point, and iterates converge to that point everywhere) or *loxodromic* (two fixed points, one attracting and one repelling). Moreover, the stabilizer of a point that contains a loxodromic element is virtually cyclic. These classical facts are [13, Lemma 6.2 and Prop. 6.4]; they apply because B is a nondegenerate continuum.

Lemma 5.5 (Parabolic comparison)**.** *Let $h \in \Gamma$ be parabolic with unique fixed point $p \in B$. Then $h^n \to p$ and $h^{-n} \to p$ in X as $n \to +\infty$; both sequences of maps converge to the constant map p uniformly on compact subsets of $X \setminus \{p\}$ and pointwise on B. Furthermore,*

$$\rho(g, gp) \longrightarrow 0 \qquad (g \to \infty \text{ in } \Gamma). \tag{5.5}$$

Proof. Convergence of powers. The powers h^n escape every finite subset of Γ as $n \to \pm\infty$ (a finite set contains finitely many powers of the infinite-order h). If a subsequence of h^n converges to $z \in B$, then by Lemma 5.1 $\rho(h^n, h^{n+1}) \to 0$ along that subsequence, while continuity of left-multiplication by h on X gives $h^{n+1} \to hz$. So $hz = z$, and uniqueness of the parabolic fixed point forces $z = p$. Compactness of X and the argument for both signs give convergence of the entire sequences.

Let $x \in X \setminus \{p\}$ and let W be a small ball about p disjoint from a compact neighborhood of x. Then $h^{-N}W$ misses x for large N (else $x = h^{-N}y$ with $y \in W$, contradicting that $h^N x = y \to p$ requires $x = p$). So $h^n x \to p$ for any such x, and compact-uniformity follows from compactness.

Comparison (5.5). Suppose the assertion fails for a sequence $g_n \to \infty$. Passing to a subsequence, we can arrange $g_n \to a \in B$ and $g_n p \to c \neq a$.

First, the cosets $g_n\langle h\rangle$ escape every finite subset of Γ. If some w were in infinitely many of them, then $g_n = wh^{j_n}$ along a subsequence with $|j_n| \to \infty$, and both $g_n \to wp$ and $g_n p \to wp$ (since $h^{\pm j_n}$ acts almost constantly on points near p for large $|j_n|$), contradiction.

For fixed n, the sequence $u_{n,j} = g_n h^j$, $j \geq 0$, starts at g_n and tends to $g_n p$. By the whole-coset escape and Lemma 5.1,

$$\sup_{j \geq 0} \rho(u_{n,j}, u_{n,j+1}) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Choose three distinct numbers strictly between 0 and $\rho(a, c)$. For each number, choose the first index $j_n^{(i)}$ at which $\rho(a, u_{n,j_n^{(i)}})$ reaches it. Vanishing of the step size plus compactness gives (after a common subsequence) three selected vertex sequences with distinct limits $z_1, z_2, z_3 \in B$.

The conjugate $k_n = g_n h g_n^{-1}$ sends each selected vertex to its successor, so asymptotically it fixes the three points z_i. If k_n escapes finite sets, pass to a subsequence with $k_n \to a'$ and $k_n^{-1} \to b'$; at least two of the z_i avoid b', and Proposition 5.4 sends both to a', contradicting distinctness. If k_n stays in a finite set (after passing to a subsequence), the k_n converge (as group elements) to a fixed conjugate of h, and by continuity that element fixes all three z_i; but h is parabolic, so its conjugate has only one fixed point. Contradiction. This proves (5.5). □

Proposition 5.6 (Minimality). *The action of Γ on B is minimal: every orbit is dense.*

Proof. Step 1. Every invariant subset with at least two points is dense. Given $a \in B$, choose group vertices $g_n \to a$ and pass to a subsequence with $g_n^{-1} \to b \in B$. An invariant subset with at least two points contains some $z \neq b$; by Proposition 5.4, $g_n z \to a$. Since $\{g_n z\} \subset$ the invariant subset, its closure contains a. Since a was arbitrary, the invariant subset is dense.

Step 2 (no global fixed point). Suppose some $c \in B$ is fixed by every element of Γ. Choose $a \neq c$ and vertices $g_n \to a$. Passing to a subsequence, Proposition 5.4 plus $g_n c = c$ forces $g_n^{-1} \to c$ (because the source triple involving c must go to some target with the same entry c, which is the repelling limit).

Choose disjoint nonempty closed neighborhoods A, C of a, c in B. For large n, $g_n A \subset A$ and $g_n^{-1} C \subset C$ (by compact-uniform convergence). Fix one such nonidentity g_n. It has infinite order (torsion-free). It cannot be parabolic: Lemma 5.5 would make the forward orbit of a point of A and the backward orbit of a point of C both tend to the same fixed point, which would then belong to both disjoint closed sets. So it is loxodromic.

Since Γ fixes c, the stabilizer of c is all of Γ and contains a loxodromic element; the classical fact quoted above then makes Γ virtually cyclic. An infinite virtually cyclic group has two ends, contradicting Lemma 3.5. So no global fixed point exists. □

6 Dimension and cohomology of the boundary

Two topological facts about B will drive the contradiction: its covering dimension is at most one, and its first Cech cohomology vanishes. Neither requires local connectivity. The dimension bound comes from small accumulation sets of coordinate cuts; the cohomology vanishing transfers a finite-cover cocycle to the presentation complex and cancels its finite coboundary there.

Thin walls and covering dimension

For a coordinate $F \in \mathcal{F}$ and a regular level t, let $P(F, t)$ be the set of endpoints of edges in $S(F, t)$, and put

$$E(F, t) = \overline{P(F, t)} \cap B.$$

In words: $E(F, t)$ is the part of B near which the cut $S(F, t)$ accumulates. All closures and diameters use the metric ρ on X.

Lemma 6.1 (Walls have measure zero)**.** *For every coordinate $F \in \mathcal{F}$, the compact set $E(F,t)$ has $\mathcal{H}^1_\rho$-measure zero for almost every $t \in (0,1)$; here $\mathcal{H}^1_\rho$ is one-dimensional Hausdorff measure with respect to ρ.*

Proof. Base coordinates. Let F be a base coordinate. The level-length estimate of Lemma 5.2, with g the identity, gives

$$\sum_{e \in S(F,t)} \sum_{e' \text{ near } e} \lambda(e') < \infty$$

for almost every t.

Fix such a level. Traversing cell boundaries along a simple dual path shows that the diameter of the endpoint set of a dual component D is at most

$$C \sum_{e \in D} \sum_{e' \text{ near } e} \lambda(e'),$$

where C depends only on the finite presentation. Bounded local incidence bounds the multiplicity with which a nearby edge is counted.

The wall has finitely many dual components, and the dual adjacency graph is locally finite. Delete finite subsets $D_n \subset S(F,t)$ increasing to the entire wall: at each stage only finitely many dual components remain, because deleting finitely many vertices from a connected locally finite graph creates only finitely many components. The endpoint-set closures of these components cover $E(F,t)$: deleting finitely many isolated graph vertices does not change boundary accumulation. The sum of their diameters is at most

$$C \sum_{e \in S(F,t) \setminus D_n} \sum_{e' \text{ near } e} \lambda(e') \longrightarrow 0.$$

So the maximum diameter tends to zero, and these finite covers prove $\mathcal{H}^1_\rho(E(F,t)) = 0$.

Localized coordinates. A localized coordinate's straddling set is contained in a finite union of base-coordinate walls at the shifted levels appearing in Lemma 4.6. The boundary accumulation set is therefore contained in the finite union of their accumulation sets. Each shifted level is admissible for almost every original level; translations and reflections of a level parameter preserve null sets. The result follows. □

Proposition 6.2 (Covering dimension at most one)**.** *The boundary B has covering dimension $\dim B \leq 1$: it admits arbitrarily fine finite open covers of multiplicity at most two.*

Proof. Choose a finite grid of levels in finitely many coordinates, each level avoiding vertex values and satisfying Lemma 6.1. Let $E = \bigcup E(F_i, t_i)$ be the union of the corresponding wall accumulation sets. It is compact and has $\mathcal{H}^1_\rho$-measure zero.

E has small clopen partitions. Fix $x \in E$. The distance map $y \mapsto \rho(x,y)$ is Lipschitz, so its image on E has Lebesgue measure zero. Hence there are arbitrarily small positive radii r with the sphere of radius r about x disjoint from E. The sets $E \cap B_\rho(x,r)$ are then clopen (both closed and open) in E. Compactness gives a finite small clopen cover, which becomes a disjoint partition by successively subtracting earlier members.

Sign tuples on $B \setminus E$. Each selected cut has a locally constant sign on the complement of its accumulation set. To define it at $x \notin E(F,t)$: choose a neighborhood W of x in X containing no endpoint of any straddling edge. Lemma 4.8 joins all vertices sufficiently near x by paths with vertices in W. Along such a path the sign of $F - t$ cannot change. Hence all sufficiently near

vertices have the same sign; that sign also applies at all sufficiently nearby boundary points, so it is locally constant on B. This definition is valid even when $F(x) = t$: it uses the signs on graph vertices approaching x.

Small diameter. The finitely many sign tuples partition $B \setminus E$ into finitely many disjoint sets open in B. Within any one of them, each tested coordinate belongs to the closure of one grid interval. First choose enough coordinates that the remaining tail of the metric $\rho = \sum_j c_j |F_j(x) - F_j(y)|$ is small, then choose the grids sufficiently fine. Every sign-tuple set consequently has arbitrarily small diameter.

Assembling the cover. Partition E into finitely many small clopen sets. They are compact, disjoint subsets of B, so they admit pairwise disjoint open neighborhoods in B, still of arbitrarily small diameter. These neighborhoods, together with the sign-tuple sets, cover B. At each point at most one of these pieces occurs (the sign-tuple sets are disjoint from each other and from a neighborhood of E near each point of E): so the cover has multiplicity at most two. A Lebesgue number for any prescribed finite open cover shows that a sufficiently fine cover of this form refines it. This is exactly $\dim B \leq 1$. □

Cech cohomology vanishes

The vanishing argument uses only the near-path property, the shrinking diameters of cell boundaries at infinity, and the group-cohomological vanishing from Lemma 3.4. It does *not* replace Cech cohomology by singular cohomology on the possibly nonlocally-connected boundary.

Proposition 6.3 (Cech cohomology vanishes)**.** *With coefficients $k = \mathbb{F}_2$ one has $\check{H}^1(B; k) = 0$.*

Proof. Let $d = (d_{ij})$ be a simplicial 1-cocycle on the nerve of a finite open cover $\{U_i\}_{i=1}^N$ of B, with $d_{ii} = 0$. We show that it is a coboundary after refinement.

Step 1 (labeling graph vertices). Choose $\delta > 0$ so that every ball of radius 3δ in B is contained in some U_i (Lebesgue number). For each vertex $u \in \Gamma$, choose a nearest point $b(u) \in B$ and a label $i(u) \in \{1, \ldots, N\}$ with $B_B(b(u), 3\delta) \subset U_{i(u)}$. If $u \to x \in B$, then $\rho(b(u), x) \leq 2\rho(u, x) \to 0$, so all labels occurring sufficiently near x contain a fixed neighborhood of x in B.

Step 2 (cellular cochain). Define a cellular 1-cochain on $\mathcal{C}$ by

$$\alpha(uv) = d_{i(u)i(v)}$$

when that pair occurs in the cover nerve, and by 0 otherwise.

Claim: the coboundary $\omega = d\alpha$ has finite support. Suppose not. Then infinitely many distinct cells have nonzero coboundary. Local finiteness of G forces their vertices to escape every finite set. Each cell boundary has at most three edges, whose lengths tend to zero (since $\lambda \in \ell^2$), so a subsequence of the vertex sets converges to one point $x \in B$. All labels on every sufficiently late cell contain x, i.e. contain a fixed neighborhood of x in some U_i. The nerve cocycle identity then gives zero around the cell boundary, contradiction. Repeated labels or boundary edges cause no difficulty: the identity is a cellular sum, and $d_{ii} = 0$.

Step 3 (group-ring cocycle). We explain why ω represents a group 2-cocycle. Start a free $k\Gamma$-resolution with the cellular chains of $\mathcal{C}$ through degree two. This is possible because $\mathcal{C}$ is simply connected: choose a free module mapping onto $\ker \partial_2$, then free modules onto successive kernels. For any finite-support scalar cellular cochain ω, the formula

$$\Phi_\omega(c) = \sum_{g \in \Gamma} \omega(g^{-1}c)\, g$$

defines a $k\Gamma$-module cochain. The sum is finite for every finite cellular chain c, and its identity coefficient is $\omega(c)$. Conversely, since the quotient has finitely many cells in each relevant degree, every module cochain through degree two is obtained this way.

For every finite cellular 2-cycle z and every $g \in \Gamma$,

$$\omega(g^{-1}z) = \alpha(\partial(g^{-1}z)) = 0$$

(since ∂ of a cycle is zero). So Φ_ω vanishes on $\ker \partial_2$ and is a cocycle on the extended free resolution. Lemma 3.4 gives $H^2(\Gamma; k\Gamma) = 0$, so there is a module 1-cochain whose coboundary is Φ_ω. Taking identity coefficients gives a finite-support scalar 1-cochain η with $d\eta = d\alpha$. Ordinary simple connectivity of $\mathcal{C}$ then supplies a scalar function l on its vertices with

$$\alpha - \eta = dl.$$

Step 4 (back to the boundary). Fix $x \in U_i$. Choose a neighborhood W of x in X avoiding all endpoints of edges in the support of η, and small enough that every vertex label occurring in W contains x. If uv is an edge with vertices in W, the sets $U_i, U_{i(u)}, U_{i(v)}$ meet at x. The cocycle identity and the equation $dl = \alpha$ on this edge give

$$l(u) + d_{i,i(u)} = l(v) + d_{i,i(v)}.$$

By Lemma 4.8, this expression has one constant value on all vertices sufficiently near x; call it $l_i(x)$. It is independent of the chosen neighborhood, since graph vertices approach every boundary point.

Step 5 (locally constant). The function l_i is locally constant on U_i: a smaller neighborhood on whose graph vertices the displayed expression is constant also computes the same germ at every sufficiently nearby boundary point. On an overlap $U_i \cap U_j$ the cocycle identity gives

$$l_i + l_j = d_{ij}.$$

Refine the cover by the open sets $U_i \cap l_i^{-1}(a)$ for $a \in k$, omitting empty ones. On this finite refinement the original cocycle is the coboundary of the assignment a to the corresponding cover member. Its Cech class vanishes. □

We have shown $\dim B \leq 1$ and $\check{H}^1(B; k) = 0$. These alone do not ensure local connectivity. The next steps isolate a continuum $L \subset B$ without cut points carrying the same properties, and then prove local connectivity by pushing graph paths toward the boundary.

7 The tree of global cut points

We now isolate the argument that uses cut points of B. Throughout this section we only use:

(Γ0) B is a nondegenerate compact metric continuum;

(Γ1) Γ is finitely presented, torsion-free, one-ended;

(Γ2) Γ acts on B minimally by convergence;

(Γ3) Γ has no nontrivial splitting over a finite or two-ended subgroup.

All these are established (Sections 3, 5). We will not use the construction of B again except through these properties.

Cut points and the pretree

A point $p \in B$ is a *global cut point* if $B \setminus \{p\}$ is disconnected. Let P be the set of global cut points. For distinct $a, p, b \in P$, write apb when a and b lie on opposite sides of an open separation of $B \setminus \{p\}$.

The relation apb is a *pretree*: a strict betweenness relation satisfying

$$\neg(aba), \qquad abc \iff cba, \qquad \neg(abc \wedge acb), \qquad abc,\ d \neq b \implies abd \text{ or } cbd.$$

Write $(a, b) = \{p : apb\}$ and $[a, b] = (a, b) \cup \{a, b\}$, with $[a, a] = \{a\}$. A subset is *full* if it contains $[a, b]$ whenever it contains a and b. A pretree is *median* if every triple has a unique point in the three pairwise closed intervals (called the *median*). It is *complete* if every nonempty full linearly ordered subset is an interval, allowed to be open or half-open.

Bowditch's theory of pretrees applies: there is a complete median pretree Φ containing P equivariantly with the same betweenness, together with a specific map and various structural results. We use the following precise statement of that theory; we cite the specific locations in [13] and prove the one additional fact that is essential for the all-parabolic case.

Proposition 7.1 (Bowditch's pretree package)**.** *Let a group act minimally by convergence on a nondegenerate compact metric continuum B, and let $P \neq \varnothing$ be its set of global cut points. Then there is a complete median pretree Φ containing P equivariantly, with the same betweenness on P, satisfying:*

(i) Adjacency. *Distinct $a, b \in \Phi$ are adjacent if $(a, b) = \varnothing$. Every adjacent pair in Φ has one point in P and one outside. A full subset whose intervals are finite is the vertex set of a simplicial tree, with these adjacent pairs as edges.*

(ii) Support. *There is an equivariant map $\phi : B \to \Phi$, equal to the identity on P, and for every $q \in \Phi$ an equivariantly assigned nonempty compact subset $R_B(q) \subset B$. For $q \in P$, $R_B(q) = \{q\}$; for $q \notin P$,*

$$R_B(q) = \phi^{-1}(q) \cup \Lambda(q), \qquad \Lambda(q) = \{p \in P : p \text{ adjacent to } q\}.$$

No point of Φ is fixed by all of Γ.

(iii) Dendritic quotient. *Beginning with equality on Φ, successively collapse finite-interval classes (two points lie in one class if their closed interval in the current quotient is finite), and at limit ordinals take the union of the preceding equivalence relations. This process stabilizes. Its final quotient is a dendrite $D(B)$; there is a continuous equivariant surjection $B \to D(B)$, and the induced action is minimal and convergence.*

(iv) Fixed-point restrictions. *If a loxodromic element fixes $q, r \in \Phi$ with $[q, r]$ infinite, then q, r are terminal (no point lies strictly between and neither is between two others). If a parabolic element has its unique fixed point in $B \setminus P$, then it has exactly one fixed point in Φ.*

Reference. The complete median pretree Φ is the flow completion of P from [13, Thm. 3.19]. Adjacency and the simplicial-tree description are [13, Lemmas 3.28 and 3.34]. The compact supports are constructed in [13, Sec. 5]; their formula is [13, Lemma 6.8], and absence of a global fixed point is [13, Lemma 6.11]. Full equivalence relations and their successive finite-interval collapses are in [13, Lemmas 4.2–4.4], and the dendritic quotient identification is in [13, Sec. 5 and Thm. 5.23]. The induced action is [13, Lemma 6.5]; minimality follows from equivariance and surjectivity. Fixed-point assertions are [13, Lemmas 6.9 and 6.10]. All these constructions apply to a continuum before local connectivity is known. □

All cut points are parabolic

The next result uses the splitting obstruction to show that the parabolic case is the only one that can occur, and that the dendritic quotient is a point.

Lemma 7.2 (Cut-point stabilizers are parabolic)**.** *Under the assumptions* (Γ0)–(Γ3)*, the dendritic quotient $D(B)$ is a single point. Every $p \in P$ has infinite stabilizer containing no loxodromic element, and hence contains a parabolic element of infinite order fixing p.*

Proof. Bowditch's dendrite obstruction [14, Lemma 1.4] applies to a finitely presented one-ended group with no infinite torsion subgroup acting minimally by convergence on a dendrite, provided every two-ended subgroup over which the group splits is parabolic on that dendrite. Here there are no such splittings, so the last condition is vacuous, and torsion-freeness implies the torsion condition. So $D(B)$ is a point.

By [13, Thm. 6.1], a minimal convergence action of a finitely generated one-ended group with no infinite torsion subgroup on a continuum has a nontrivial dendritic quotient whenever there is a cut point which is not parabolic. Its contrapositive shows every point of P is parabolic in the sense that its stabilizer is infinite and contains no loxodromic element. Any nonidentity element of this stabilizer has infinite order because Γ is torsion-free; by the convergence action classification it is therefore parabolic. □

The cited Theorem 6.1 does not itself produce an invariant finite-interval class when all cut points are parabolic. We now prove that, adapting the limit and higher-successor steps of Bowditch's proof following his Lemma 6.13. The parabolic fixed-point observation below permits those steps.

Lemma 7.3 (Parabolic fixed points in Φ)**.** *Let γ be parabolic with unique fixed point $a \in P$. Every point of Φ fixed by γ is either a or adjacent to a. Consequently, if an infinite-order element fixes two points of Φ whose interval is infinite, the element is loxodromic and both points are terminal.*

Proof. Every nonempty compact γ-invariant subset of B contains a: the iterates of any other point converge to a (Lemma 5.5). If $q \in \Phi$ is fixed by γ, then $R_B(q) \subset B$ is such a compact invariant subset, so contains a. For $q \in P$ this gives $q = a$ (unique fixed point in B). For $q \notin P$: $a \in R_B(q)$ and $\phi(a) = a \neq q$, so by the formula of Proposition 7.1(ii) we must have $a \in \Lambda(q)$, i.e. q is adjacent to a.

Any two points equal to or adjacent to a have a finite interval between them, contained in the union of their intervals to a. So a parabolic element fixing a point of P cannot have two fixed points with an infinite interval between them. A parabolic element whose fixed point lies outside P cannot have them either by Proposition 7.1(iv). The infinite-order element that has such a pair of fixed points is therefore loxodromic, and by Proposition 7.1(iv) both fixed points are terminal. □

An invariant finite-interval class

The final quotient $D(B)$ is a point. We now show an invariant class already appears at the first finite-interval collapse.

An equivalence relation on Φ is *full* if all its classes are full. For a full relation $\sim$, the quotient pretree has the induced betweenness: a class V lies between distinct classes U, W when some $v \in V$ lies between every $u \in U$ and every $w \in W$. Write $\sim_0$ for equality and define

$$x \sim_{\beta+1} y \iff [x,y] \text{ meets finitely many } \sim_\beta\text{-classes}, \qquad \sim_\lambda = \bigcup_{\beta<\lambda} \sim_\beta \ \ (\lambda \text{ a limit ordinal}). \tag{7.1}$$

These are the increasing invariant full relations from Proposition 7.1(iii); quotients remain complete median pretrees. In particular $x \sim_1 y$ means $[x, y]$ is finite. Goal: a fixed point of $\Phi/\sim_1$.

We record a fact from completeness that will control the direction of an edge between collapsed classes. Let U, V be disjoint nonempty full subsets of a complete median pretree with full union. A *facing point of V toward U* is a point $p \in V$ such that

$$[u, v] \cap V = [p, v] \qquad (u \in U, v \in V). \tag{7.2}$$

Lemma 7.4 (Facing point exists and is unique)**.** *There is at most one facing point of V toward U on each side, and at least one of U, V admits one.*

Proof. Choose $u \in U$, $v \in V$. Partition $[u, v]$ by the two full sets. The initial part $A = [u, v] \cap U$ has, by completeness, the form $[u, t]$ or $[u, t)$.

Closed case. $t \in U$. Then t is the "last point of U" on $[u, v]$, and the point immediately adjacent (or the endpoint of the adjacent open interval, if V extends into $[u, v]$) witnesses the facing point either in U or in V.

Half-open case. Put $m = \mathrm{med}(u, v, t)$. If $m \neq t$ then $A = [u, m]$, reducing to the closed case with m in place of t. Otherwise $t \in [u, v]$, $t \notin A$, so $t \in V$. Then t is the facing point of V.

This proves existence. Uniqueness: if $p, p' \in V$ both work, then $\mathrm{med}(u, p, p')$ must equal both p and p' (via (7.2)), forcing $p = p'$. Interchanging U and V shows the symmetric statement. This is also [13, Lemma 6.13]. □

Lemma 7.5 (An invariant finite-interval class)**.** *Under* (Γ0)–(Γ3)*, Γ preserves a finite-interval class of Φ.*

Proof. Lemma 7.2 says the terminal quotient in (7.1) is a point. Let α be the least ordinal for which $\Phi/\sim_\alpha$ has a point fixed by Γ. Proposition 7.1(ii) gives $\alpha \neq 0$. We rule out limits and successors greater than one.

Limit case. Suppose α is a limit ordinal. Choose x in an invariant $\sim_\alpha$-class. For each member s of a finite generating set of Γ, the relation $x \sim_\alpha sx$ holds at some stage $\beta_s < \alpha$. Let $\beta = \max_s \beta_s < \alpha$. Then invariance and transitivity of $\sim_\beta$ give $x \sim_\beta gx$ for every word g in the generators and their inverses, i.e. the $\sim_\beta$-class of x is invariant, contradicting minimality of α.

Higher successor case. Suppose $\alpha = \beta + 1$ with $\beta \geq 1$. Let $\Xi \subset \Phi$ be an invariant $\sim_\alpha$-class. Its $\sim_\beta$-classes form a simplicial tree $\Sigma = \Xi/\sim_\beta$. The action on this tree has no fixed vertex by minimality of α.

For adjacent classes $U, V \in \Sigma$, their union is full. So by Lemma 7.4 there is a facing point on at least one side. There cannot be facing points on both sides: they would be adjacent in Φ, hence $\sim_1$-equivalent and therefore $\sim_\beta$-equivalent, contradicting $U \neq V$. So exactly one side has a facing point, and an automorphism cannot interchange U, V. Thus the action on Σ has no edge inversions.

Every edge with infinite stabilizer is a leaf edge. Let UV be an edge of Σ with infinite stabilizer, with $p \in V$ its facing point (say; all direction labels below are interchangeable). Choose a nonidentity stabilizing element γ; it has infinite order (torsion-free). γ preserves U and V and fixes p.

Claim: γ fixes a second point outside V. For $y \in U$ put $x = \mathrm{med}(p, y, \gamma y)$. Fullness gives $x \in [y, \gamma y] \subset U$; both x and γx lie in the linearly ordered interval $[p, \gamma y]$. If $x = \gamma x$, done. Otherwise, replacing (γ, x) by $(\gamma^{-1}, \gamma x)$ if necessary, we get $x \in (p, \gamma x)$. The intervals $[p, \gamma^n x]$,

$n \geq 0$, are then strictly increasing. Their union is a full directed arc with first point p and no last point. Completeness gives an endpoint z with

$$\bigcup_{n\geq 0} [p, \gamma^n x] = [p, z). \tag{7.3}$$

The endpoint is unique in a median pretree [13, Lemma 3.13]. The union in (7.3) is γ-invariant, so z is fixed. Since $x \in (p, z)$ but $x \notin V$, fullness of V forces $z \notin V$.

So γ fixes p and a second point $q \notin V$. The points p, q belong to q in different $\sim_\beta$-classes (since $\beta \geq 1$, $\sim_\beta$-inequality implies infinite original interval). By Lemma 7.3, p is terminal in Φ. Now for $v \in V$ and $u \in U$, by facing we have $p \in [u, v]$. Since $p \neq u$, terminality forces $v = p$. So $V = \{p\}$, i.e. V is a leaf vertex of Σ (if it had two neighbors, p would lie strictly between representatives of them in Φ, contradicting terminality).

Delete leaf edges and get an invariant subtree. Delete every edge with infinite stabilizer, together with its leaf endpoint. The remaining vertices form a nonempty invariant subtree (removing selected leaves preserves paths between remaining vertices). If nothing remained, the original tree would have at most two vertices, so an action without inversions would have a fixed vertex, already excluded. All remaining edges have finite stabilizer. A fixed vertex of the remaining tree would also be fixed in Σ.

But a finitely generated one-ended group acting without inversions on a simplicial tree with finite edge stabilizers fixes a vertex [13, Lemma 6.6]. Contradiction.

The least ordinal is therefore $\alpha = 1$, which is exactly the assertion of the lemma. □

The simplicial tree

Proposition 7.6 (Tree of cut points)**.** *Let* (Γ0)–(Γ3) *hold and suppose $P \neq \varnothing$. Then there is a nontrivial minimal simplicial Γ-tree $\mathcal{T}$ with no edge inversions and finite quotient, such that:*

(i) *P is an invariant subset of its vertices, and every edge has one endpoint in P and one outside;*

(ii) *$\mathcal{T}$ is spanned by P; its other vertices have degree at least two;*

(iii) *every edge stabilizer is infinite;*

(iv) *if $a, b, q \in P$ and q separates a and b in B, then $q \in [a, b]_{\mathcal{T}}$.*

Every point of P is fixed by an infinite-order parabolic element.

Proof. Let $C \subset \Phi$ be the invariant finite-interval class supplied by Lemma 7.5. It has more than one point (else Φ would have a fixed point, excluded by Proposition 7.1(ii)). So its simplicial tree has an edge, and Proposition 7.1(i) places one endpoint of that edge in P.

$P \subset C$. Choose $p \in C \cap P$. The orbit Γp is dense in B (Proposition 5.6) and lies in C. Now let $q \in P$. Take an open separation of $B \setminus \{q\}$. Both sides are nonempty open subsets, so each meets Γp. For orbit points a, b on opposite sides, separation gives $q \in [a, b]_\Phi$. Since C is full, $q \in C$.

$\mathcal{T}$ spanned by P. Let $\mathcal{T}$ be the subtree of C spanned by P (union of intervals between points). Its intervals agree with those of Φ, so it retains the separation compatibility (iv). Adjacency in Proposition 7.1(i) gives the bipartition P, and forbids edge inversions. A vertex outside P cannot be a leaf of a tree spanned by P.

Minimality. Let $\mathcal{S} \subset \mathcal{T}$ nonempty invariant subtree. A single vertex would be a fixed point of Φ, excluded. Otherwise $\mathcal{S}$ contains an edge, hence a point of P. Its orbit is again dense; separation gives $P \subset \mathcal{S}$. So $\mathcal{S} = \mathcal{T}$.

Finite quotient. Choose a base vertex o and a finite generating set S for Γ. The union of the finite paths $[o, so]$, $s \in S$, is a finite subtree after adjoining o. Its translates form a connected invariant subtree, hence cover $\mathcal{T}$ by minimality. So the quotient has finitely many vertex and edge orbits.

Infinite edge stabilizers. If an edge had finite stabilizer, collapsing all edges outside its orbit gives a nontrivial tree with one edge orbit and a minimal action without inversions. Bass-Serre theory yields a nontrivial splitting over that edge stabilizer, contrary to (Γ3).

The parabolic assertion is Lemma 7.2. □

8 The subcontinuum without cut points

We have seen that B always has $\dim \leq 1$ and $\check{H}^1 = 0$ (Propositions 6.2, 6.3). We now show that B always contains a *nondegenerate subcontinuum L without cut points* that retains these two properties, is a retract of B, and comes equipped with the graph-theoretic structure needed for path pushing.

If B has no cut points, we just take $L = B$; the case is simple. Otherwise we use the tree $\mathcal{T}$ from Section 7 to extract one of its "leaf blocks".

The setup

Assume $P \neq \varnothing$ for the rest of this subsection. The tree $\mathcal{T}$ from Proposition 7.6 is bipartite between P and the remaining vertices; every edge has a P-endpoint and a non-P-endpoint. Choose *any* vertex $r \in \mathcal{T} \setminus P$ and set

$$R = \{p \in P : p \text{ is adjacent to } r \text{ in } \mathcal{T}\}, \qquad H = \Gamma(r) \leq \Gamma.$$

Here $\Gamma(r)$ is the stabilizer of r.

Since $\mathcal{T}$ is spanned by P, a non-P vertex cannot be a leaf, so $|R| \geq 2$. For distinct $a, b \in R$, the tree interval is $a - r - b$. By Proposition 7.6(iv), no cut point of B separates a and b.

Also H is exactly the setwise stabilizer of R: any element preserving R preserves the unique midpoint r of the interval $a - r - b$ for any two distinct $a, b \in R$.

A graph on the cut set

Lemma 8.1 (Graph on P)**.** *There is a connected Γ-graph K^* with vertex set P and finitely many edge orbits such that the endpoints of every edge have tree-distance two in $\mathcal{T}$. The graph K obtained by restricting K^* to R is connected with finitely many H-orbits of vertices and edges.*

Define the projection $\pi : P \to R$ by: $\pi(p)$ is the neighbor of r on the tree interval from r to p. Then an edge of K^* with both endpoints in R is fixed by π, and every other edge is collapsed to a vertex. Moreover the stabilizer in H of every vertex of K is infinite.

Proof. Constructing K^*. The finite quotient of $\mathcal{T}$ (Proposition 7.6) gives finitely many Γ-orbits in P. Choose representatives $p_0, \ldots, p_k$ of P-orbits and a finite symmetric generating set S of Γ. Start with the graph containing all translates of the edges $p_0 \to sp_0$ for $s \in S$, and of $p_0 \to p_i$ for $1 \leq i \leq k$ (omit loops, i.e. when $p_0 = sp_0$).

This graph is connected: the s-edges connect the whole p_0-orbit, and the $p_0 \to p_i$-edges connect every other orbit to p_0. It has finitely many edge orbits.

Replace each edge by the *consecutive pairs* of P-vertices along the tree interval between its endpoints. These tree intervals are finite (they are between P-points in $\mathcal{T}$). The replacement is

equivariant. So the resulting K^* is still connected, has finitely many edge orbits, and the endpoints of every edge are at tree-distance two.

Projection properties. Let asb be an edge of K^* with midpoint s (so a, s, b are collinear in $\mathcal{T}$ and s is the midpoint of the interval a-b). Two cases:

- $s = r$: then $a, b \in R$ and $\pi(a) = a$, $\pi(b) = b$ — the edge is fixed by π.
- $s \neq r$: then a and b lie in the same component of $\mathcal{T} \setminus \{r\}$, so they have the same neighbor on the path to r, i.e. $\pi(a) = \pi(b)$ — the edge collapses.

This shows: projecting a path between two members of R and deleting stationary steps gives a path in K, so K is connected.

Orbits and stabilizers. If a group element carries one edge of K to another, it carries their unique tree midpoints (both equal to r) to one another, so it lies in H. Consequently every Γ-orbit of K^*-edges meets K in at most one H-orbit, so there are finitely many H-edge orbits in K. Since K is connected with at least two vertices, it has finitely many H-vertex orbits.

Finally, for $p \in R$: $H_p \subseteq H$ is the stabilizer of the tree edge rp in the whole group Γ, which is infinite by Proposition 7.6(iii). □

The graph K can have infinite valence and infinite vertex stabilizers. This will cause no trouble: our arguments use orbit sets, not finite valence.

Lemma 8.2 (Paths in K near B)**.** *Suppose $a_n, b_n \in P$ both tend to $x \in B$. Then there are paths in K^* from a_n to b_n whose vertices all tend uniformly to x. If a path in K^* has distinct projected endpoint labels in R, then after stationary steps are removed, its projection uses only vertices of the original path.*

Proof. By Lemma 8.1 (finitely many representatives), write $a_n = g_n p_{i_n}$ and $b_n = h_n p_{j_n}$ with $i_n, j_n \in \{0, \ldots, k\}$. We may choose g_n, h_n to escape every finite subset of Γ: each representative p_i has infinite stabilizer H_{p_i} (since p_i is a vertex of P and each has infinite stabilizer by Proposition 7.6), so we can multiply by any element of that stabilizer.

By Proposition 7.6 every P-point is a parabolic fixed point; a parabolic fixed point has $\rho(gp_i, g) \to 0$ as $g \to \infty$ (Lemma 5.5). Applied to the finite list of possible p_i, we get $g_n \to x$ and $h_n \to x$ in X.

Now by Lemma 4.8, the Cayley paths from g_n to h_n (in G, the original graph) have all vertices tending uniformly to x. Fix finite K^*-paths from p_0 to sp_0 for every Cayley generator s, and from p_0 to each representative p_i. Replace each Cayley step $g \to gs$ by the g-translate of the first appropriate K^*-path, and similarly at the endpoints (using the translated paths to p_{i_n} and p_{j_n}). The result is a K^*-path from a_n to b_n.

The fixed K^*-path templates involve only finitely many points of P. For each such point t, we have $\rho(gt, g) \to 0$ as g escapes finite sets (parabolic comparison). The Cayley vertices escape finite sets uniformly because they tend to the boundary point x. So all transferred path vertices tend uniformly to x.

Regarding the last assertion: by Lemma 8.1, every non-constant projected edge (i.e. an edge whose image under π is not a collapsed vertex) was already an edge of the original path. So when the projected endpoints differ, every vertex of the projected path lies on the projection of a path edge in the original path. □

Retraction onto the subcontinuum

Proposition 8.3 (Retraction and the subcontinuum L)**.** *Assume $P \neq \varnothing$. The map $\pi : P \to R$ extends to a continuous retraction*

$$\mathfrak{r} : B \to L, \qquad L = \overline{R}^{B}.$$

The map $\mathfrak{r}$ is locally constant on $B \setminus L$ and is the identity on L. Moreover:

$$L \text{ is a nondegenerate continuum without cut points}, \qquad \dim L \leq 1, \qquad \check{H}^1(L; \mathbb{F}_2) = 0.$$

Proof. Minimality gives density. By Proposition 5.6, the invariant nonempty set P is dense in B.

π extends continuously on $B \setminus L$. Let $x \in B \setminus L$. Choose a neighborhood O of x in B with closure disjoint from L. Claim: π is constant on $P \cap V$ for some neighborhood $V \subseteq O$ of x. If not, we could choose $a_n, b_n \in P \to x$ with $\pi(a_n) \neq \pi(b_n)$. Lemma 8.2 would join a_n to b_n by paths whose vertices eventually lie entirely in O. Any such path with distinct projected endpoints uses at least one vertex of R lying in O (by the last part of that lemma). But $O \subseteq B \setminus L$ and $R \subseteq L$, so $O \cap R = \varnothing$ — contradiction.

π extends continuously on L. Now let $x \in L$ and $a_n \in P \to x$. Choose $b_n \in R \to x$ (density of R in L). Join a_n to b_n by paths from Lemma 8.2. Whenever $\pi(a_n) \neq b_n$, the same lemma places $\pi(a_n)$ on the original path, so $\rho(\pi(a_n), b_n) \to 0$, hence $\pi(a_n) \to x$. On the other hand, if $\pi(a_n) = b_n$, then $\pi(a_n) = b_n \to x$ directly. So in all cases $\pi(a_n) \to x$.

Formally constructing $\mathfrak{r}$. Take the closure of the graph of π in $B \times L$. By density of P and compactness of L, this closed set projects onto B. The two paragraphs above show it is a graph: exactly one point of L over each point of B. Its projection onto B is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. The second coordinate defines $\mathfrak{r} : B \to L$.

By the first paragraph, $\mathfrak{r}$ is locally constant on $B \setminus L$. It's the identity on L by construction (the graph of π contains (q, q) for every $q \in R$, and by density and continuity it contains (q, q) for every $q \in L$).

Continuum properties. B is connected and L is its continuous image, so L is connected. R has at least two points, so L is nondegenerate. Since L is closed in the compact metric space B, it is compact metric, and $\dim L \leq \dim B \leq 1$.

Čech cohomology of L. Let $i : L \hookrightarrow B$ be the inclusion. Then $\mathfrak{r} \circ i = \mathrm{id}_L$. Functoriality of Čech cohomology gives $i^* \mathfrak{r}^* = \mathrm{id}_{\check{H}^1(L;\mathbb{F}_2)}$. So $\mathfrak{r}^*$ injects $\check{H}^1(L; \mathbb{F}_2)$ into $\check{H}^1(B; \mathbb{F}_2) = 0$. Thus $\check{H}^1(L; \mathbb{F}_2) = 0$.

L has no cut points. Suppose toward contradiction that $q \in L$ is a cut point of L: $L \setminus \{q\} = U \sqcup V$ with U, V nonempty open in L. By density, there is $a \in R \cap U$ and $b \in R \cap V$. The extra fiber

$$F = \mathfrak{r}^{-1}(\{q\}) \setminus \{q\}$$

is closed in $B \setminus \{q\}$ by continuity of $\mathfrak{r}$. It is also open in $B \setminus \{q\}$: it lies in $B \setminus L$, where $\mathfrak{r}$ is locally constant, hence near any point of F the map $\mathfrak{r}$ is constant on a neighborhood; since $\mathfrak{r}$ takes value q at that point of F, the whole neighborhood is in F. Therefore

$$B \setminus \{q\} = \big(\mathfrak{r}^{-1}(U) \cup F\big) \sqcup \mathfrak{r}^{-1}(V)$$

is a separation: $\mathfrak{r}^{-1}(U)$, $\mathfrak{r}^{-1}(V)$ are open (by continuity), F is open (as above) and disjoint from both, and $a \in \mathfrak{r}^{-1}(U)$, $b \in \mathfrak{r}^{-1}(V)$ lie on opposite sides. This would make q a cut point of B separating two members of R, contradicting the observation (from Proposition 7.6(iv)) that no cut point of B separates two distinct members of R. □

The common setting for path pushing

We now include the case $P = \varnothing$ and state the common setting that the path-pushing argument uses. In the $P = \varnothing$ case we just take $L = B$ and the original graph.

Proposition 8.4 (Setting for path pushing)**.** *There are a subgroup $H \leq \Gamma$, a compact space $Y = H \sqcup L$ (with the metric ρ inherited from X), and a connected H-graph K with vertex set $I \subseteq Y$, such that:*

(1) *$Y = \overline{H}^X$; the points of H are isolated in Y; L is a nondegenerate continuum without cut points satisfying $\dim L \leq 1$ and $\check{H}^1(L; \mathbb{F}_2) = 0$.*

(2) *If $h_n \to a \in L$ and $h_n^{-1} \to b \in L$, then h_n converges to the constant map a, uniformly on compact subsets of $Y \setminus \{b\}$.*

(3) *K has finitely many H-orbits of vertices and edges; either $I = H$ or $I \subseteq L$; every point of L is the limit of a sequence of distinct vertices of I.*

(4) *For each fixed $w \in I$: $\rho(hw, h) \to 0$ as $h \to \infty$ in H.*

(5) *For every $x \in L$ and every neighborhood W of x in Y, there is a neighborhood V of x in Y such that any two vertices of $I \cap V$ can be joined by a finite path of K with all vertices in W.*

(6) *If $I \subseteq L$, then for every $p \in I$ there is an infinite-order element $h \in H_p$ with $h^j z \to p$ for every $z \in L$ as $j \to +\infty$.*

No finite-generation of H and no local finiteness of K is assumed.

Proof. Case $P = \varnothing$. Take $H = \Gamma$, $Y = X$, $L = B$, $K =$ the Cayley graph G with $I = H$. Continuum properties are those of B (Lemma 4.7; absence of cut points is the assumption of this case). (2) is Proposition 5.4. (3), (4) are from the construction of X, Proposition 5.4, Lemma 4.8, and Lemma 5.1. Every boundary point is approached by distinct group vertices because isolated vertices accumulate on B everywhere (dense in X). (5) is Lemma 4.8. (6) has no content: $I = H$ means I is disjoint from L.

Case $P \neq \varnothing$. Take

$$H = \Gamma(r), \qquad I = R, \qquad L = \overline{R}^B, \qquad Y = H \sqcup L,$$

and K the graph of Lemma 8.1. Property (1) is Lemma 4.7, Proposition 8.3. For the remaining interfaces:

$Y = \overline{H}^X$: fix $p \in R$. If an escaping sequence $h_n \in H$ converges to a boundary point, then parabolic comparison ($\rho(h, hp) \to 0$) and $h_n p \in R$ show that the limit lies in L. Conversely, for any $p \in R$, the infinite stabilizer H_p supplies distinct h_n with $h_n = p$ (i.e. $h_n p = p$, but the h_n themselves are distinct and tend to p by parabolic comparison). So $R \subseteq \overline{H}^X$, and taking closures gives $L \subseteq \overline{H}^X$. No point of $\Gamma \setminus H$ lies in the closure (group vertices are isolated). This proves equality.

Property (2): restriction of Proposition 5.4 to H.

Property (3): finite orbit sets come from Lemma 8.1. $I = R \subseteq L$ (since $R \subseteq \overline{R}^B = L$). Every point of L is a limit of distinct members of R: R is dense in L and L has no isolated points (else that point would be isolated in the continuum B, which has no isolated points).

Property (4): parabolic comparison (Lemma 5.5).

Property (5): near-path for K. Take two sequences $a_n, b_n \in R$ tending to the same point $x \in L$. Lemma 8.2 joins them by K^*-paths with vertices tending uniformly to x. Project to R: if the endpoints of the projected path differ, the projected path uses only vertices of the original path, so it still tends uniformly to x. If the endpoints of the projection agree, use the constant path (at that vertex). So the projected paths give the required K-paths. In a metric space, this sequential formulation gives the neighborhood formulation.

Property (6): H_p is infinite by Lemma 8.1. Torsion-freeness of Γ makes it contain an infinite-order element. By Lemma 7.2, that element is parabolic with fixed point p. Its powers act as asserted by Lemma 5.5. □

9 Annuli, component labels, and depth

The goal of the next two sections is to prove that the continuum L from Proposition 8.4 is locally connected. We do this by pushing graph paths toward the boundary. The two new tools are: *component labels* on L (which record which graph component a given boundary point's near-vertices belong to), and *annular depth* (a numerical depth that measures how far an annulus extends from a fixed set).

Standing assumptions from Proposition 8.4: we have a compact metric space $Y = H \sqcup L$ (with metric ρ), a group H acting on Y by homeomorphisms extending left multiplication, H-points isolated, and L a nondegenerate continuum without cut points satisfying $\dim L \leq 1$ and $\check{H}^1(L; k) = 0$, where $k = \mathbb{F}_2$. The action satisfies the convergence action property (2). We also have a connected H-graph K with vertex set $I \subseteq Y$, where $I = H$ or $I \subseteq L$, with finitely many H-orbits of vertices and edges, satisfying the near-path property (5), the comparison (4), and, in the $I \subseteq L$ case, the stabilizer property (6).

No finite generation of H and no local finiteness of K is assumed.

Extending nearby labels across a deleted point

Suppose a locally constant map takes a small punctured neighborhood $V \setminus \{x\}$ of a boundary point x to some discrete label set. When does it extend across x? If $\check{H}^1(L; k) = 0$ and x is not a cut point, the answer is: always (for k-valued labels). The circle shows the difference absence of cut points alone makes.

Lemma 9.1 (Extension across a non-cut point)**.** *Let L be a nondegenerate compact metric continuum without cut points and with $\check{H}^1(L; k) = 0$. If $x \in L$ and V is an open neighborhood of x in L, every locally constant map $u : V \setminus \{x\} \to k$ extends uniquely to a locally constant map on V.*

Proof. Put $U = L \setminus \{x\}$. We prove via finite-cover cohomology that there are locally constant $f_U : U \to k$ and $f_V : V \to k$ with $u = f_U + f_V$ on $U \cap V$.

Choose open sets U_0, V_0 covering L with $\overline{U_0} \subseteq U$ and $\overline{V_0} \subseteq V$. The set $C = \overline{U_0} \cap \overline{V_0}$ is compact in $U \cap V$. Its two subsets $C \cap u^{-1}(0)$ and $C \cap u^{-1}(1)$ are disjoint compact. Choose $\delta > 0$ smaller than their distance (any δ if either is empty).

Take a finite open cover (W_i) of L refining (U_0, V_0), with each W_i of diameter $< \delta$, and give it a parent $P_i \in \{U, V\}$ according to this refinement. On every nonempty $W_i \cap W_j$, define $c_{ij} = 0$ if $P_i = P_j$, and $c_{ij} = u$ value otherwise. The latter is a constant: the intersection lies in C and cannot meet both of C's two compact pieces. At a triple intersection either all parents agree (so all three values are 0) or exactly two are from one parent and one from the other (so the two transition values on the two sides equal the same u-value). Thus (c_{ij}) is a 1-cocycle on the nerve.

Since $\check{H}^1(L;k) = 0$, some finite open refinement (Z_α) with refinement map r admits constants $a_\alpha \in k$ satisfying $a_\alpha + a_\beta = c_{r(\alpha)r(\beta)}$ whenever $Z_\alpha \cap Z_\beta \neq \varnothing$.

Define f_U on $U \cap Z_\alpha$: if $P_{r(\alpha)} = U$, set $f_U = a_\alpha$; otherwise set $f_U = a_\alpha + u$. The displayed identity makes these definitions agree on overlaps; they are locally constant. Similarly for f_V (with roles of U, V reversed). Then $f_U + f_V = u$ on $U \cap V$.

Since x is not a cut point, U is connected, so f_U takes one constant value, say c. Then $f_V + c$ is the required extension. Uniqueness: a nondegenerate continuum has no isolated points, so $V \setminus \{x\}$ is dense in V; two continuous functions agreeing on a dense set agree everywhere. □

Component labels

For a subset $D \subset Y$, let D' be the set of limits of sequences of distinct points of D. In a metric space, D' is closed. For an allowed vertex set $T \subseteq I$, put

$$\Omega_T = L \setminus (I \setminus T)'.$$

This is exactly the open set of boundary points near which *deleted vertices do not accumulate as distinct points*. We use components of the induced graph $K[T]$.

Lemma 9.2 (Component labels)**.** *Within the setting above, each $x \in \Omega_T$ has a unique component $Q_T(x)$ of $K[T]$ such that all allowed vertices in some neighborhood of x belong to $Q_T(x)$. Every $x \in \Omega_T$ is a limit of distinct allowed vertices, and the assignment Q_T is locally constant on Ω_T. If $t \in T \cap L \cap \Omega_T$ (i.e. t is both an allowed vertex and a limit of distinct allowed vertices in the whole space), then $Q_T(t)$ is the component containing t itself.*

Proof. There is a neighborhood W of x such that $(I \setminus T) \cap W \subseteq \{x\}$. Since x is a limit of distinct vertices of I (Proposition 8.4(3)), it is consequently a limit of distinct vertices of T as well.

Case 1: x is not a deleted vertex. After shrinking W, it contains no deleted vertex. The near-path property (5) then puts all vertices sufficiently near x into one component of $K[T]$. There are allowed vertices arbitrarily near x, so this component is unique. The same neighborhood, together with approximation by allowed vertices, shows the component assignment is locally constant near x. If x itself is an allowed vertex, it belongs to this component.

Case 2: x is a deleted vertex (so $I \subseteq L$ and $x \in I \setminus T$). Choose an open neighborhood V of x in L containing no other deleted vertex. By Case 1, every $y \in V \setminus \{x\}$ has a component label $q(y)$, and these form a locally constant map

$$q : V \setminus \{x\} \to \{\text{components of } K[T]\}.$$

Compose any map from the label set $\to k$ with q to get a map $V \setminus \{x\} \to k$. Each such map extends across x by Lemma 9.1. We claim this forces q itself to be constant sufficiently near x.

Call a label *recurrent* if it occurs arbitrarily near x.

- If two labels are recurrent, color one 0 and the other 1; that coloring separates them, contradicting extension (a locally constant map on $V \setminus \{x\}$ extending across x cannot take both values on recurrent labels).
- If exactly one label is recurrent but q is not eventually constant, color the recurrent label 0 and all others 1, again contradicting extension.
- If no label is recurrent, choose a sequence of points tending to x with pairwise distinct labels (at each step, exclude the finitely many labels already chosen). Color the successive labels alternately 0 and 1. This again contradicts extension as a locally constant map.

Thus one component label occurs throughout a small punctured neighborhood of x. Every allowed vertex there belongs to its own label by Case 1, so all these vertices belong to that component.

This proves existence at a deleted vertex too. Approximation by allowed vertices proves uniqueness and, on shrinking the neighborhood once more, local constancy of Q_T there. □

The annulus system

An *annulus* in Y is an ordered pair $\sigma = (\sigma^-, \sigma^+)$ of disjoint closed subsets of Y such that each side meets L and $\sigma^- \cup \sigma^+ \neq Y$. The sides need not be connected. For annuli σ, τ, write

$$\sigma < \tau \iff \sigma^+ \cup \tau^- = Y.$$

Then $\sigma^- \subseteq \operatorname{int}(\tau^-)$ and $\tau^+ \subseteq \operatorname{int}(\sigma^+)$. The relation is transitive and irreflexive: a nested chain $\sigma_1 < \cdots < \sigma_n$ uses distinct annuli. A chain *goes from F to E* if $F \subseteq \sigma_1^-$ and $E \subseteq \sigma_n^+$.

Lemma 9.3 (Annulus system)**.** *There is a set $\mathcal{A}$ of annuli invariant under H and under interchange of the two sides, with:*

1. *For every $\delta > 0$, only finitely many annuli in $\mathcal{A}$ have both sides of diameter at least δ.*
2. *For every closed $F \subseteq Y$, every $x \in L \setminus F$, and every integer $n \geq 1$, there is a chain of n annuli in $\mathcal{A}$ from F to some neighborhood of x in Y.*

Proof. Uniform convergence is uniform. For every $r, \varepsilon > 0$, all but finitely many $g \in H$ admit a point $b_g \in L$ with

$$\operatorname{diam} g\big(Y \setminus B_\rho(b_g, r)\big) < \varepsilon.$$

Otherwise choose distinct exceptions g_n and pass to a subsequence with $g_n \to a \in L$ and $g_n^{-1} \to b \in L$. Uniform convergence on the compact $Y \setminus B_\rho(b, r)$ (property (2)) gives a contradiction.

Choose $r_i \downarrow 0$ with $r_i < \operatorname{diam}(L)/3$ and $\varepsilon_i = 2^{-i}$. Apply the observation with $r = r_i/4$ and $\varepsilon = \varepsilon_i$, gaining a finite exceptional set $E_i \subset H$. Uniform continuity of the finitely many maps in E_i allows $0 < s_i < r_i/4$ with

$$\operatorname{diam} g\big(\overline{B_\rho(y, s_i)}\big) < \varepsilon_i \quad (g \in E_i, y \in L).$$

Choose finitely many centers $y_{ij} \in L$ with $s_i/2$-balls covering L. At each center take the seed annulus

$$\sigma_{ij}^- = Y \setminus B_\rho(y_{ij}, r_i), \qquad \sigma_{ij}^+ = \overline{B_\rho(y_{ij}, s_i)}.$$

Its sides are disjoint with distance $\geq r_i - s_i$. The positive side contains its center. The negative side meets L: every point of L is at distance $\geq \operatorname{diam}(L)/2$ from some point of L (extreme points of a diameter). The sides cannot cover L: their nonempty closed intersections with L would disconnect it.

Let $\mathcal{A}$ consist of all distinct translates of these annuli and their reversals.

Diameter bound. A ball of radius $r_i/4$ cannot meet both sides of a seed annulus (whose mutual distance exceeds $3r_i/4$). For $g \notin E_i$, one side lies outside the exceptional ball for g and has image of diameter $< \varepsilon_i$. For $g \in E_i$, the positive side has that property by choice of s_i. Hence every translated level-i annulus has at least one side of diameter $< \varepsilon_i$.

Now fix a seed and $\delta > 0$. Only finitely many distinct translates have both sides of diameter $\geq \delta$: otherwise choose such corresponding distinct group elements and a convergence subsequence with repeller b. At least one of the two closed seed sides excludes b, so its images have diameter

tending to zero. There are only finitely many seeds at each level, and levels with $\varepsilon_i < \delta$ contribute no annuli with two such large sides. Hence property (1).

Chain property. Let F be closed and $x \in L \setminus F$. At a sufficiently fine level, choose a center y_{ij} with $\rho(x, y_{ij}) < s_i/2$ and $r_i + s_i/2 < \operatorname{dist}(x, F)$ when F is nonempty. The negative side contains F, and the positive side contains a neighborhood of x. Suppose an annulus already chosen has its positive side containing $B_\rho(x, \eta)$. Choose a finer level with $r_j + s_j/2 < \eta$ and a center y within $s_j/2$ of x. Then $B_\rho(y, r_j) \subseteq B_\rho(x, \eta)$, so the new annulus is *larger* in the order $<$, and its positive side again contains a neighborhood of x. Repeating gives any prescribed finite length. □

Depth

Fix such an annulus system $\mathcal{A}$. For $F \subseteq Y$ and $v \in Y$, define

$$D(F, v) = \sup\{n \geq 1 : \text{a chain of length } n \text{ in } \mathcal{A} \text{ goes from } F \text{ to } \{v\}\},$$

with $D = 0$ if no such chain exists. In general $D \in \{0, 1, 2, \ldots\} \cup \{\infty\}$. Write $D(u, v)$ for $D(\{u\}, v)$. This function counts separating annuli; it is not a metric.

Lemma 9.4 (Properties of depth)**.** *For $g \in H$, $F \subseteq Y$, and $v \in Y$,*

$$D(gF, gv) = D(F, v).$$

If F contains a point of H and $v \in H$, then $D(F, v)$ is finite. When it is positive, a chain attains this maximum. If F is closed, $x \in L \setminus F$, and $N \geq 1$, then there is a neighborhood V_N of x in Y with $D(F, v) \geq N$ for every $v \in V_N$.

Proof. Equivariance. Translation carries annuli and chains bijectively to annuli and chains.

Finiteness. Suppose $h \in F \cap H$ and $v \in H$. Since L is closed and does not contain h or v (points of H are not in L), the number $\delta = \min\{\operatorname{dist}(h, L), \operatorname{dist}(v, L)\}$ is positive. Throughout a chain from F to v: every negative side contains h and meets L, every positive side contains v and meets L. So both sides have diameter $\geq \delta$. Lemma 9.3(1) supplies only finitely many such annuli, and a strict chain cannot repeat an annulus. So $D(F, v)$ is finite.

Attainment. A finite nonzero supremum of possible integer lengths is attained (the set of achievable lengths is a downward-closed subset of positive integers, so its supremum is one of its members).

Deep neighborhoods. The final claim is exactly Lemma 9.3(2) applied to length N, using the resulting terminal neighborhood of x. □

10 Pushing paths and local connectedness

We work in the setting of Proposition 8.4: a compact metric space $Y = H \sqcup L$, an H-graph K on $I \subseteq Y$, and the annulus system $\mathcal{A}$ from Lemma 9.3. Our goal: prove L is locally connected.

The idea: fix a closed set A containing 1 and avoiding L, and introduce a depth-based subgraph $K[I_m]$ that captures "vertices deeper than m away from A". Component labels for $K[I_m]$ give a locally constant map Q_m on $L \setminus A$. The key uniformity is: beyond one threshold M, increasing m to $m + 1$ never separates two boundary points that shared a label at level m. Then paths in $K[I_m]$ joining nearby vertices persist to arbitrary depth, and their Hausdorff limits give connected subsets of L of small diameter.

Uniform depth comparison

Fix a closed subset $A \subset Y$ with $1 \in A$ and $L \setminus A \neq \varnothing$. Write

$$d_0 = \operatorname{dist}(1, L) > 0, \qquad N(a, b) = \#\{\sigma \in \mathcal{A} : \operatorname{diam} \sigma^- \geq a,\ \operatorname{diam} \sigma^+ \geq b\},$$

which is finite by Lemma 9.3(1). Any side containing 1 has diameter $\geq d_0$ (every side meets L).

Lemma 10.1 (Shrinking toward the boundary)**.** *Let $g_n \in H$ satisfy $D(A, g_n) \to \infty$ and set $F_n = g_n^{-1} A$. Along a subsequence there is $q \in L$ such that F_n converges in the Hausdorff metric to $\{q\}$.*

Proof. By equivariance, $D(F_n, 1) = D(A, g_n)$. In any chain from F_n to 1, every positive side contains 1. If the chain has more than $N(\varepsilon, d_0)$ members, one of its negative sides has diameter $< \varepsilon$; that side contains F_n, so $\operatorname{diam} F_n < \varepsilon$. Since ε was arbitrary, $\operatorname{diam} F_n \to 0$.

Lemma 9.4 gives $D(A, g) < \infty$ for every fixed $g \in H$, so g_n escapes every finite set (as does g_n^{-1}). Since $1 \in A$, we have $g_n^{-1} \in F_n$. Compactness of Y and isolatedness of its group vertices give a subsequence $g_n^{-1} \to q \in L$. Diameter conclusion closes it. □

Lemma 10.2 (Uniform depth comparison)**.** *Suppose $F_n = g_n^{-1} A$ converges to $\{q\}$ as in Lemma 10.1. For each $r > 0$ set*

$$C_r = N(r/2, d_0) + N(d_0, r) + 2.$$

For all sufficiently large n, uniformly over every $b \in H$ with $\rho(b, q) \geq 3r$,

$$D(F_n, b) \geq D(F_n, 1) + D(\{1\}, b) - C_r. \tag{10.1}$$

Consequently, if $b_n \in H \to z \in L \setminus \{q\}$, then $D(F_n, b_n) \geq D(F_n, 1) + D(\{1\}, b_n) - C$ for a constant C, and $D(\{1\}, b_n) \to \infty$.

Proof. Prune the initial chain. Take n large enough that $F_n \subseteq B(q, r/2)$ and $\ell = D(F_n, 1) > N(r/2, d_0) + 1$. Take a chain $\sigma_1 < \cdots < \sigma_\ell$ of maximum length from F_n to 1, and put $U = B(q, r)$.

If $\sigma_j^- \not\subset U$, its negative side contains a point of F_n and a point outside U, so has diameter $\geq r/2$. Its positive side has diameter $\geq d_0$. There are at most $N_- = N(r/2, d_0)$ such annuli. Because negative sides increase along a chain, those form a terminal segment. Remove that segment and one further annulus. The retained initial chain has $\geq \ell - N_- - 1$ members. The negative side of the next annulus lies in U, so the last retained positive side contains $Y \setminus U$. In particular, the retained chain already ends on the side containing any b under consideration.

Prune the tail chain. Let $t = D(\{1\}, b)$ and take a maximum chain $\tau_1 < \cdots < \tau_t$ from 1 to b (when $t > 0$). Put $V = Y \setminus \overline{B(q, 2r)}$. A positive side not contained in V contains both b and a point at distance $\leq 2r$ from q, hence diameter $\geq r$. Every negative side contains 1, hence diameter $\geq d_0$. So at most $N_+ = N(d_0, r)$ positive sides fail to lie in V; those form an initial segment. If $t > N_+ + 1$, remove this initial segment and one further annulus. The retained final chain has $\geq t - N_+ - 1$ members, and its first negative side contains $Y \setminus V$.

Concatenate. Since $U \cap V = \varnothing$, the last positive side of the first retained chain and the first negative side of the second cover Y. The defining order relation therefore concatenates the two chains. Their total length is at least $\ell + t - N_- - N_+ - 2$. If $t \leq N_+ + 1$, the first retained chain alone has at least this length. This proves (10.1) with a constant and threshold independent of b.

Final claim. Choose r with $4r < \rho(q, z)$. Eventually $\rho(b_n, q) \geq 3r$, so the inequality applies. For any prescribed integer N, Lemma 9.3(2) gives a chain of length N from $\{1\}$ to a neighborhood of z. Eventually this neighborhood contains b_n, so $D(\{1\}, b_n) \to \infty$. □

Components of deeper vertex sets

Choose one representative p_i for each H-orbit in I; when $I = H$, use only $p_1 = 1$. For u in the orbit of p_i, define its set of representatives $C(u) = \{g \in H : gp_i = u\}$. For integers $m \geq 0$ define the *deep vertex set*

$$I_m = \left\{ u \in I \setminus A : \min_{g \in C(u)} D(A, g) \geq m \right\}. \tag{10.2}$$

Each depth in the minimum is a finite nonnegative integer (by Lemma 9.4: A contains $1 \in H$, so there is a finiteness associated). The condition asks about *every* representative; this will let us slide through an infinite stabilizer without losing the depth bound.

Lemma 10.3 (Component labels for deep vertices)**.** *For each fixed m, deleted vertices $I \setminus I_m$ have no accumulation by distinct vertices at any point of $L \setminus A$. Consequently Lemma 9.2 defines a locally constant map*

$$Q_m : L \setminus A \to \{\textit{components of } K[I_m]\},$$

whose value at x contains all vertices of I_m sufficiently near x. Every $x \in L \setminus A$ is approached by distinct vertices with this label. For every finite $E \subset I$, one has $E \cap I_m = \varnothing$ for all sufficiently large m.

Proof. No accumulation. For $m = 0$, the assertion follows from $I_0 = I \setminus A$. Suppose $m \geq 1$ and distinct $u_n \in I \setminus I_m \to x \in L \setminus A$. Eventually $u_n \notin A$, so choose $a_n \in C(u_n)$ with $D(A, a_n) < m$. Since there are only finitely many orbit representatives and u_n are distinct, a_n escapes every finite set. The comparison $\rho(gp_i, g) \to 0$ (Proposition 8.4(4)) applied to the finite list of representatives gives $a_n \to x$. A chain from A to a neighborhood of x of length m (Lemma 9.3(2)) contradicts $D(A, a_n) < m$ for large n.

Structure of Q_m. The claim about Q_m and approximation follows from Lemma 9.2.

Eventually excluded. For $u \in E \setminus A$, choose one representative $a_u \in C(u)$. The depth $D(A, a_u)$ is finite. Taking m larger than all these finitely many depths excludes every $u \in E$ from I_m. □

Lemma 10.4 (Translated deletions stay bounded)**.** *Let $m_n \to \infty$ and $D(A, g_n) \geq m_n$. Pass to a subsequence with $F_n = g_n^{-1}A \to \{q\}$, $q \in L$. For every compact $E \subset Y \setminus \{q\}$ there is a finite $S_E \subset I$ and an index n_E with*

$$E \cap g_n^{-1}(I \setminus I_{m_n+1}) \subseteq S_E \qquad (n \geq n_E).$$

Proof. Supposing not, choose indices $\to \infty$ and pairwise distinct $w_n \in E \cap g_n^{-1}(I \setminus I_{m_n+1})$. Compactness gives $w_n \to z \in E$. The limit lies in L: automatic if $I \subseteq L$; if $I = H$, follow from isolatedness of group vertices. Since $F_n \to q \notin E$, eventually $w_n \notin F_n$. The deletion of $g_n w_n$ gives a representative $g_n b_n \in C(g_n w_n)$ with $D(F_n, b_n) = D(A, g_n b_n) < m_n + 1$. After restricting to one orbit representative, $w_n = b_n p_i$. Distinctness forces b_n to escape finite sets, and comparison gives $b_n \to z \neq q$. Lemma 10.2 then yields

$$D(F_n, b_n) \geq m_n + D(\{1\}, b_n) - C > m_n + 1$$

for large n, contradiction. □

Lemma 10.5 (Eventual labels)**.** *For sequences as in Lemma 10.4, define $\ell_n(z) = Q_{m_n+1}(g_n z)$ whenever $z \in L$ and $g_n z \notin A$. For each $z \in L \setminus \{q\}$ there are a relative neighborhood $V_z \subset L \setminus \{q\}$ and an index n_z such that all values $\ell_n(w)$ with $w \in V_z$, $n \geq n_z$ are defined and $\ell_n(w) = \ell_n(z)$. For any two points $z, z' \in L \setminus \{q\}$, $\ell_n(z) = \ell_n(z')$ for all sufficiently large n.*

Proof. Choose a neighborhood O of z in Y whose closure misses q. Apply Lemma 10.4 to the compact $E = \overline{O}$. There are finite $S \subset I$ and index n_0 with $T = (I \cap O) \setminus S$ satisfying $g_n T \subseteq I_{m_n+1}$ for all $n \geq n_0$. The deleted vertices $I \setminus T$ have no distinct accumulation at any point of $O \cap L$. Lemma 9.2 gives a neighborhood $V_z \subset O \cap L$ on which the component of $K[T]$ approached by vertices is constant; call it C. Shrink V_z if needed. Since $F_n \to \{q\}$, all $g_n w$ for $w \in V_z$ lie outside A once n is large.

Fix such an n. Every $w \in V_z$ is approached by vertices of C; their images approach $g_n w$ and lie in the connected subgraph $g_n C$ of $K[I_{m_n+1}]$. The defining property of Q_{m_n+1} identifies $\ell_n(w)$ with the component containing $g_n C$, for every $w \in V_z$. That gives the asserted neighborhood and threshold.

Equivalence and single class. On $L \setminus \{q\}$, declare $z \sim z'$ if their labels agree for all sufficiently large n. Eventual equality is an equivalence relation, even though the label set may change with n. The first part shows each equivalence class is open. Since q is not a cut point, $L \setminus \{q\}$ is connected, so there is one class. □

A common threshold

We use three fixed boundary points to get a single threshold M that works for every graph path simultaneously.

Choose a set $Z \subset L$ of three distinct points (exists because L is a nondegenerate continuum).

Lemma 10.6 (Majority labels)**.** *For every finite $S \subset H$, there is $M = M(A, Z, S) \geq 1$ such that for all integers $m \geq M$:*

(i) If $D(A, g) \geq m$, at least two points of gZ lie in $L \setminus A$ and have a common Q_{m+1} label. Call it the majority label *$J_m(g)$.*

(ii) If $s \in S$ and $D(A, g), D(A, gs) \geq m$, then $J_m(g) = J_m(gs)$.

Proof. Part (i). If the first assertion fails at arbitrarily large depths, choose $m_n \to \infty$ and g_n violating it with $D(A, g_n) \geq m_n$. Pass to a subsequence with $g_n^{-1} A \to \{q\}$. Two fixed points of Z avoid q. By Lemma 10.5, their images under g_n eventually lie outside A and share a single Q_{m_n+1} label, contradiction. Uniqueness of the majority label also follows: two different labels cannot each occur at two points of a three-point set.

Part (ii). After choosing a threshold for (i), suppose (ii) fails at arbitrarily large m. Choose violating sequences and, since S is finite, restrict to one fixed s. Again pass to a subsequence with $g_n^{-1} A \to \{q\}$. Choose two points of $Z \setminus \{q\}$ and two points of $sZ \setminus \{q\}$. Lemma 10.5 makes the labels of all four images under g_n equal for large n. They determine the majority labels of both $g_n Z$ and $g_n sZ$, contradicting the chosen inequality. □

Fixing the finite set S. There are finitely many orbits of ordered edges of K: allowing both orientations, at most twice the number of edge orbits. For each representative (u_e, v_e) choose $a_e, b_e \in H$ and indices $i(e), j(e)$ with $u_e = a_e p_{i(e)}$ and $v_e = b_e p_{j(e)}$, and include $s_e = a_e^{-1} b_e$ in S. Then the endpoints of every edge have representatives g and gs for some $s \in S$.

When $I \subseteq L$, also include, for each p_i, a stabilizing element $h_i \in H$ from Proposition 8.4(6) with $h_i p_i = p_i$ and $h_i^j z \to p_i$ for $z \in L$ as $j \to +\infty$.

Fix the threshold M from Lemma 10.6 for this list.

Lemma 10.7 (Refinement does not split labels)**.** *For every $m \geq M$ and all $x, y \in L \setminus A$,*

$$Q_m(x) = Q_m(y) \implies Q_{m+1}(x) = Q_{m+1}(y). \tag{10.3}$$

In particular, M is independent of x and y.

Proof. Case 1: $I \subseteq L$. Claim: for $v \in I_m$ and $g \in C(v)$,

$$J_m(g) = Q_{m+1}(v). \tag{10.4}$$

Let $v = gp_i$. For every $j \geq 0$, $gh_i^j \in C(v)$. The minimum in (10.2) gives $D(A, gh_i^j) \geq m$ for every j. The move $h_i \in S$ and Lemma 10.6 give

$$J_m(g) = J_m(gh_i) = J_m(gh_i^2) = \cdots .$$

As $j \to \infty$, the three points $gh_i^j Z$ tend to v. Since $v \in L \setminus A$ and Q_{m+1} is locally constant there, all three anchors eventually have label $Q_{m+1}(v)$. This proves (10.4).

Suppose $v, w \in I_m$ are adjacent. Choose g, gs representatives via the finite edge list. Both depths $\geq m$ by the minimum convention. Lemma 10.6 and (10.4) give $Q_{m+1}(v) = Q_{m+1}(w)$. So Q_{m+1} is constant on every path in $K[I_m]$. If $Q_m(x) = Q_m(y)$, choose vertices in a common component sufficiently near x and y (using Lemma 10.3 and local constancy of Q_{m+1}); a path in that component proves (10.3).

Case 2: $I = H$. Only one representative $p_1 = 1$, and $C(v) = \{v\}$. The finite edge moves and Lemma 10.6 imply J_m is constant along paths of $K[I_m]$. For $x \in L \setminus A$ and $m \geq M$ fixed, claim that $J_m(g) = Q_{m+1}(x)$ for all group vertices g sufficiently near x. These vertices lie in I_m when sufficiently near x by Lemma 10.3. If the claim fails, take $g_n \to x$ with the wrong label and pass to a subsequence with $g_n^{-1} \to b \in L$. The convergence property (2) sends at least two points of $Z \setminus \{b\}$ to x under g_n. Local constancy of Q_{m+1} near x gives the asserted majority label, contradiction. Approximating x, y inside their common Q_m component and using constancy of J_m along a connecting path proves (10.3). □

Limits of pushed paths

Proposition 10.8 (Local connectivity of L)**.** *In the setting of Proposition 8.4, the continuum L is locally connected.*

Proof. Fix $x \in L$ and a relatively open neighborhood $U \subset L$ of x. Choose an open neighborhood O of x in Y with $1 \notin O$ and $\overline{O} \cap L \subset U$, and apply the preceding construction to $A = Y \setminus O$.

Let M be the threshold of Lemma 10.7. By local constancy of Q_M, there is a neighborhood $V \subset L \setminus A$ of x with $Q_M(y) = Q_M(x)$ for all $y \in V$. Induction via Lemma 10.7 gives

$$Q_m(y) = Q_m(x) \qquad (y \in V,\ m \geq M).$$

Fix $y \in V$. For each $m \geq M$, choose $a_m, b_m \in I_m$ in the common label component with $\rho(a_m, x) < 1/m$ and $\rho(b_m, y) < 1/m$, and choose P_m a finite path in $K[I_m]$ joining them. Every vertex of P_m lies in O.

Mesh tends to zero. Use the finite list of edge representatives (u_e, v_e) and the comparison ($\rho(gu_e, gv_e) \to 0$ uniformly over the list as g escapes). For each $\varepsilon > 0$, choose a finite $F \subset H$ outside which every edge distance is $< \varepsilon$. Only finitely many endpoints of edges (gu_e, gv_e) with $g \in F$ exist. By Lemma 10.3, none of those belongs to I_m once m is large. So every edge in $K[I_m]$ has endpoint distance $< \varepsilon$. This proves the mesh assertion with no local-finiteness assumption on K.

Hausdorff limit is connected. Regard P_m as its finite vertex set. Nonempty closed subsets of the compact metric space Y are compact in the Hausdorff metric; pass to a subsequence $P_m \to C$.

The limit contains x, y and lies in $\overline{O}$. It also lies in L: immediate if $I \subseteq L$; if $I = H$, Lemma 10.3 excludes any fixed group vertex from P_m for large m, and isolatedness of that vertex excludes it from the limit.

If C were disconnected, write it as a disjoint union of nonempty compact C_0, C_1 with distance $\delta > 0$. For large m, P_m lies in the disjoint union of $\delta/3$-neighborhoods of C_0, C_1 and meets both. A path in these vertices then has two consecutive vertices in different neighborhoods, at distance $\geq \delta/3$. This contradicts the mesh assertion. So C is connected.

Conclusion. C is a connected subset of $\overline{O} \cap L \subseteq U$ containing x and y. So every $y \in V$ lies in the same component of U as x. Repeating at every point and every open subset of L shows that components of open sets in L are open. This is local connectivity. □

11 The continuum contradiction

We now have: a nondegenerate compact metric continuum L that is locally connected, has $\dim L \leq 1$, and has $\check{H}^1(L; \mathbb{F}_2) = 0$. This section extracts a contradiction.

The idea: locally connected + $\dim \leq 1$ + no circle forces L to be a dendrite; every nondegenerate dendrite has a cut point; but L has none. Contradiction.

Two lemmas we need

Lemma 11.1 (Cohomology of closed subsets)**.** *If Z is a compact metric space with $\dim Z \leq 1$, $A \subset Z$ closed, and k is an abelian coefficient group, then restriction induces a surjection*

$$\check{H}^1(Z; k) \to \check{H}^1(A; k).$$

Proof. Represent a class on A by a cocycle d on the nerve of a finite relative open cover $\{V_i\}$ of A. Choose open sets $W_i \subset Z$ with $W_i \cap A = V_i$. The cover $\{W_i\} \cup \{Z \setminus A\}$ of Z has a finite open refinement $\mathcal{U}$ of multiplicity at most two (that's $\dim Z \leq 1$). Its nerve $N(\mathcal{U})$ is therefore a graph.

Restrict $\mathcal{U}$ to A: keep the indices of the cover members, even if two restrictions coincide. After empty restrictions are discarded, the nerve $N(\mathcal{U}|_A)$ is a subgraph of $N(\mathcal{U})$. A refinement map pulls the original cocycle d back to a 1-cochain on this subgraph.

Extend that 1-cochain by zero on every remaining edge of $N(\mathcal{U})$. Since $N(\mathcal{U})$ is a graph (a 1-dimensional complex, no 2-simplices), the extension is automatically a 1-cocycle. Its Čech class in $\check{H}^1(Z; k)$ restricts to the prescribed class on A. □

Lemma 11.2 (Path-connected open subsets)**.** *Let L be a locally connected compact metric space with metric d, and let $U \subset L$ be connected and open. Then any two points $a, b \in U$ are joined by a continuous path in U.*

Proof. Cover by small connected pieces. Every connected open $V \subset L$ is covered by connected open sets W whose closures lie in V and whose diameter is at most any prescribed $\varepsilon > 0$: around any point $x \in V$, take a small metric ball with closure in V, then a connected open neighborhood inside a still smaller ball (local connectivity supplies it).

Connected intersection graph. Take the cover from the previous paragraph. Its intersection graph is connected: otherwise, the union of members in a graph component would be a nonempty open and relatively closed subset of V, separating V.

Chain of overlapping sets. Any two points $a, b \in U$ can be joined by a finite chain $W_1, \ldots, W_m$ from this cover, with $a \in W_1$, $b \in W_m$, and $W_i \cap W_{i+1} \neq \varnothing$. We may arrange $m \geq 2$ by repeating a member.

Iterative subdivision. Starting with $[0,1]$, U, and endpoints a, b, construct finite interval partitions $\mathcal{P}_n$ as follows. For each $I \in \mathcal{P}_n$, associate a connected open set V_I containing its already-assigned endpoint values, with $\operatorname{diam} V_I \le 2^{-n}$ (for $n \ge 1$). Inside V_I, take the finite chain above with closures in V_I and diameters $\le 2^{-(n+1)}$. Subdivide I into m equal pieces and associate these to the chain members. At each new division point assign a point of the corresponding consecutive intersection. Keep the old endpoint values. Successive assignments agree at shared endpoints, and both endpoints of every child interval lie in its associated set.

Construction. Every subdivision has at least two children, so $\operatorname{mesh}(\mathcal{P}_n) \le 2^{-n}$. The set D of all partition endpoints is therefore dense in $[0,1]$, and the compatible assignments define $f : D \to L$. Nesting ensures $f(D \cap I) \subset V_I$ for every $I \in \mathcal{P}_n$.

Uniform continuity. For fixed $n \ge 1$, let δ_n be the minimum length of a cell of $\mathcal{P}_n$. If $s, t \in D$ and $|s - t| < \delta_n$, they lie in the same or adjacent cells. Both cases give $d(f(s), f(t)) \le 2^{1-n}$. So f is uniformly continuous and extends continuously to $[0,1]$ by completeness of L.

Path stays in U. For each cell $I \in \mathcal{P}_1$, the extended image lies in $\overline{V_I} \subset U$. The full path therefore lies in U, and its endpoints are a, b. □

The final contradiction

Lemma 11.3 (Every such continuum has a cut point)**.** *Every nondegenerate locally connected compact metric continuum L with*

$$\dim L \le 1, \qquad \check{H}^1(L; \mathbb{F}_2) = 0$$

has a cut point.

Proof. L contains no embedded circle. Suppose $C \subset L$ were an embedded circle S^1. It is closed in L. Lemma 11.1 gives a surjection $\check{H}^1(L; \mathbb{F}_2) \to \check{H}^1(C; \mathbb{F}_2)$. But the left side is zero while the right side is $\mathbb{F}_2$ (a circle has first Čech cohomology $\mathbb{F}_2$). Contradiction.

Choose an arc. A locally connected compact metric continuum is a *Peano continuum*. The theory of Peano continua (via Moore's theorem) says that in any connected open subset of L, any two points can be joined by an arc. (A capstone: continuous images of intervals contain embedded arcs, via [16, pp. 301–304]. Combined with Lemma 11.2, arcs exist wherever paths do.) Pick an arc from a to b (distinct points of L), and an interior point p of that arc.

If p were not a cut point: contradiction. The connected open set $L \setminus \{p\}$ would contain a second arc, say γ, from a to b. Let α be the component of the first arc minus γ that contains p; this is an open subarc of the first arc with both endpoints on γ. The closure of α, together with the subarc of γ joining the two endpoints of α, is an embedded circle.

But L has no embedded circle. So p must be a cut point. □

Proof of Theorem 1.2. Suppose $b_1^{(2)}(\widetilde{Q}) > 0$. By the finite-energy construction (Sections 4–5) and the boundary topology (Section 6) we have built a nondegenerate compact metric continuum B with $\dim B \le 1$ and $\check{H}^1(B; \mathbb{F}_2) = 0$.

If B has no cut points, set $L = B$; the reduction of Proposition 8.4 (with $P = \varnothing$) applies directly. Otherwise, use Section 7 and Section 8 to isolate a subcontinuum $L \subset B$ without cut points, still nondegenerate, still with $\dim L \le 1$ and $\check{H}^1(L; \mathbb{F}_2) = 0$. In both cases, Proposition 10.8 gives local connectivity of L.

Then Lemma 11.3 forces L to have a cut point, so L would be a nondegenerate locally connected compact metric continuum with a cut point — but L was constructed *without* cut points (Proposition 8.3 in the $P \neq \varnothing$ case, by definition in the $P = \varnothing$ case). Contradiction.

Therefore $b_1^{(2)}(\widetilde{Q}) = 0$. Lemma 3.7 gives $b_0^{(2)} = b_4^{(2)} = 0$, $b_3^{(2)} = b_1^{(2)} = 0$, and vanishing in all degrees above four. Its duality argument includes nontrivial orientation characters. Consequently $b_p^{(2)}(\widetilde{Q}; \mathcal{N}(\Gamma)) = 0$ for every $p \geq 0$ with $p \neq 2$. □

Proof of Corollary 1.3 (Singer in dimension four). Let M be a closed connected aspherical topological four-manifold. Lemma 3.3 supplies a finite aspherical integral Poincaré-complex homotopy model of M. A homotopy equivalence lifts to an equivariant homotopy equivalence of universal covers, so it preserves the singular-chain L^2-Betti numbers. Theorem 1.2 applies to the model. □

12 Euler characteristic, signature, and finite covers

The main theorem determines the remaining (degree-two) L^2-Betti number from a classical topological invariant. This gives constraints on Euler characteristics and signatures of closed aspherical four-manifolds.

Corollary 12.1 (Euler characteristic)**.** *Let Q be a finite connected aspherical integral Poincaré complex of formal dimension four, with arbitrary orientation character. Then*

$$\chi(Q) = b_2^{(2)}(\widetilde{Q}) \geq 0.$$

Moreover $\chi(Q) = 0$ if and only if every L^2-Betti number of $\widetilde{Q}$ vanishes. The same conclusions hold for closed connected aspherical topological four-manifolds.

Proof. The L^2 Euler-Poincaré formula for a finite CW complex is

$$\chi(Q) = \sum_{p \geq 0} (-1)^p b_p^{(2)}(\widetilde{Q})$$

[8, Theorem 1.35(2)]. By Theorem 1.2, only the degree-two term survives (all other $b_p^{(2)}$, $p \neq 2$, vanish, and those p are all we need consider since $b_p^{(2)} = 0$ for $p > 4$). Its von Neumann dimension is nonnegative. The characterization of zero follows from the same theorem. For a topological manifold, pass to the finite homotopy model of Lemma 3.3; the invariants are preserved. □

The middle term need not vanish: for closed oriented hyperbolic surfaces Σ_g, Σ_h of genera $g, h \geq 2$, the product $\Sigma_g \times \Sigma_h$ is aspherical and has $\chi = 4(g-1)(h-1) > 0$. On the other hand the four-torus is a flat aspherical four-manifold with $\chi = 0$.

For an oriented closed four-manifold M, let $\sigma(M)$ be the signature of the intersection form on $H^2(M; \mathbb{R})$ (equivalently, on $H_2(M; \mathbb{R})$ via Poincaré duality).

Corollary 12.2 (Euler-signature inequality)**.** *Every closed connected oriented aspherical topological four-manifold M satisfies*

$$\chi(M) \geq |\sigma(M)|.$$

In particular, $\chi(M) = 0$ implies $\sigma(M) = 0$.

Proof. The topological L^2-signature theorem (which does not require smoothness) gives $\sigma(M) = \sigma^{(2)}(\widetilde{M})$ [18, Theorem 0.2]. The right side is the difference of the positive and negative von Neumann dimensions of the lifted middle-dimensional intersection form, so $|\sigma^{(2)}(\widetilde{M})| \leq b_2^{(2)}(\widetilde{M})$. Corollary 12.1 now proves the inequality. □

The corollary includes non-smoothable manifolds: the cited signature theorem is topological. Its equality between ordinary and L^2 signatures for a finite Poincaré complex is an additional input; we have not assumed it here.

Corollary 12.3 (Betti growth along finite covers)**.** *Let Q be as in Corollary 12.1, and suppose $\Gamma = \pi_1(Q)$ admits a descending sequence $\Gamma = \Gamma_0 \supseteq \Gamma_1 \supseteq \cdots$ of finite-index normal subgroups with $\bigcap_i \Gamma_i = \{1\}$. Let $Q_i \to Q$ be the associated finite covers. For every $p \geq 0$,*

$$\lim_{i\to\infty} \frac{b_p(Q_i;\mathbb{Q})}{[\Gamma:\Gamma_i]} = \begin{cases} \chi(Q), & p = 2, \\ 0, & p \neq 2. \end{cases}$$

The same statement holds for a closed connected aspherical topological four-manifold and its corresponding covers.

Proof. Lück's approximation theorem [17, Theorem 0.1] identifies the limit with $b_p^{(2)}(\widetilde{Q})$. Apply Theorem 1.2 and Corollary 12.1. A finite homotopy model of a topological manifold lifts to a homotopy equivalence on each finite cover, giving the last assertion. □

13 Bibliography

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